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A-Mem: Agentic Memory for LLM Agents

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Abstract

While large language model (LLM) agents can effectively use external tools for complex real-world tasks, they require memory systems to leverage historical experiences. Current memory systems enable basic storage and retrieval but lack sophisticated memory organization, despite recent attempts to incorporate graph databases. Moreover, these systems' fixed operations and structures limit their adaptability across diverse tasks. To address this limitation, this paper proposes a novel agentic memory system for LLM agents that can dynamically organize memories in an agentic way. Following the basic principles of the Zettelkasten method, we designed our memory system to create interconnected knowledge networks through dynamic indexing and linking. When a new memory is added, we generate a comprehensive note containing multiple structured attributes, including contextual descriptions, keywords, and tags. The system then analyzes historical memories to identify relevant connections, establishing links where meaningful similarities exist. Additionally, this process enables memory evolution – as new memories are integrated, they can trigger updates to the contextual representations and attributes of existing historical memories, allowing the memory network to continuously refine its understanding. Our approach combines the structured organization principles of Zettelkasten with the flexibility of agent-driven decision making, allowing for more adaptive and context-aware memory management. Empirical experiments on six foundation models show superior improvement against existing SOTA baselines.

Code for Benchmark Evaluation:
https://github.com/WujiangXu/AgenticMemory

Code for Production-ready Agentic Memory:
https://github.com/WujiangXu/A-mem-sys

1 Introduction

Large Language Model (LLM) agents have demonstrated remarkable capabilities in various tasks, with recent advances enabling them to interact with environments, execute tasks, and make decisions autonomously [23, 33, 7]. They integrate LLMs with external tools and delicate workflows to improve reasoning and planning abilities. Though LLM agent has strong reasoning performance, it still needs a memory system to provide long-term interaction ability with the external environment [35].

Existing memory systems [25, 39, 28, 21] for LLM agents provide basic memory storage functionality. These systems require agent developers to predefine memory storage structures, specify storage points within the workflow, and establish retrieval timing. Meanwhile, to improve structured memory organization, Mem0 [8], following the principles of RAG [9, 18, 30], incorporates graph databases for storage and retrieval processes. While graph databases provide structured organization for memory systems, their reliance on predefined schemas and relationships fundamentally limits their adaptability. This limitation manifests clearly in practical scenarios - when an agent learns a novel mathematical solution, current systems can only categorize and link this information within their preset framework,

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A-MEM：代理式记忆用于大语言模型代理

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摘要

虽然大语言模型（LLM）代理可以有效地使用外部工具来处理复杂的现实世界任务，但它们需要记忆系统来利用历史经验。当前的记忆系统支持基本的存储和检索，但在引入图数据库的尝试后，仍然缺乏复杂的记忆组织能力。此外，这些系统的固定操作和结构限制了它们在不同任务中的适应性。为了解决这一局限性，本文提出了一种针对LLM代理的新型能动记忆系统，该系统能够以能动的方式动态组织记忆。遵循卡片盒方法的基本原则，我们设计了记忆系统，通过动态索引和链接创建相互连接的知识网络。当添加新记忆时，系统会生成包含多个结构化属性的全面笔记，包括上下文描述、关键词和标签。然后，系统分析历史记忆以识别相关连接，在存在有意义相似性的地方建立链接。此外，该过程还能实现记忆进化——随着新记忆的整合，它们可以触发现有历史记忆的上下文表示和属性的更新，使记忆网络能够持续完善其理解。我们的方法结合了卡片盒的结构化组织原则和代理驱动的决策灵活性，实现了更具适应性和上下文感知的记忆管理。在六个基础模型上的实证实验表明，与现有SOTA基线相比，我们的方法取得了显著的改进。

基准评估代码:
https://github.com/WujiangXu/AgenticMemory**生产就绪的代理式记忆代码**:
https://github.com/WujiangXu/A-mem-sys

1 简介

大语言模型（LLM）代理在各种任务中展现出卓越的能力，最近的进展使它们能够与环境交互、执行任务并自主决策 [23, 33, 7]。它们将LLM与外部工具和精细的工作流程相结合，以提升推理和规划能力。尽管LLM代理具有强大的推理性能，但它仍然需要一个记忆系统来提供与外部环境进行长期交互的能力 [35]。
现有的为LLM代理提供的记忆系统 [25, 39, 28, 21] 提供基础内存存储功能。这些系统需要代理开发者预先定义内存存储结构，在流程中指定存储点，并建立检索时机。与此同时，为了改进结构化记忆组织，遵循RAG [9, 18, 30], 原则的Mem0 [8], 将图数据库用于存储和检索过程。虽然图数据库为记忆系统提供了结构化组织，但它们对预定义模式和关系的依赖从根本上限制了其适应性。这一局限性在实践中表现得非常明显——当代理学习一种新的数学解决方案时，当前系统只能在其预设框架内对这一信息进行分类和链接，

预印本。

(a) Traditional memory system. **(b)** Our proposed agentic memory.

Figure 1: Traditional memory systems require predefined memory access patterns specified in the workflow, limiting their adaptability to diverse scenarios. Contrastly, our A-MEM enhances the flexibility of LLM agents by enabling dynamic memory operations.

unable to forge innovative connections or develop new organizational patterns as knowledge evolves. Such rigid structures, coupled with fixed agent workflows, severely restrict these systems' ability to generalize across new environments and maintain effectiveness in long-term interactions. The challenge becomes increasingly critical as LLM agents tackle more complex, open-ended tasks, where flexible knowledge organization and continuous adaptation are essential. Therefore, *how to design a flexible and universal memory system that supports LLM agents' long-term interactions* remains a crucial challenge.

In this paper, we introduce a novel agentic memory system, named as A-MEM, for LLM agents that enables dynamic memory structuring without relying on static, predetermined memory operations. Our approach draws inspiration from the Zettelkasten method [15, 1], a sophisticated knowledge management system that creates interconnected information networks through atomic notes and flexible linking mechanisms. Our system introduces an agentic memory architecture that enables autonomous and flexible memory management for LLM agents. For each new memory, we construct comprehensive notes, which integrates multiple representations: structured textual attributes including several attributes and embedding vectors for similarity matching. Then A-MEM analyzes the historical memory repository to establish meaningful connections based on semantic similarities and shared attributes. This integration process not only creates new links but also enables dynamic evolution when new memories are incorporated, they can trigger updates to the contextual representations of existing memories, allowing the entire memories to continuously refine and deepen its understanding over time. The contributions are summarized as:

• We present A-MEM, an agentic memory system for LLM agents that enables autonomous generation of contextual descriptions, dynamic establishment of memory connections, and intelligent evolution of existing memories based on new experiences. This system equips LLM agents with long-term interaction capabilities without requiring predetermined memory operations.

• We design an agentic memory update mechanism where new memories automatically trigger two key operations: link generation and memory evolution. Link generation automatically establishes connections between memories by identifying shared attributes and similar contextual descriptions. Memory evolution enables existing memories to dynamically adapt as new experiences are analyzed, leading to the emergence of higher-order patterns and attributes.

• We conduct comprehensive evaluations of our system using a long-term conversational dataset, comparing performance across six foundation models using six distinct evaluation metrics, demonstrating significant improvements. Moreover, we provide T-SNE visualizations to illustrate the structured organization of our agentic memory system.

2 Related Work

2.1 Memory for LLM Agents

Prior works on LLM agent memory systems have explored various mechanisms for memory management and utilization [23, 21, 8, 39]. Some approaches complete interaction storage, which maintains comprehensive historical records through dense retrieval models [39] or read-write memory structures [24]. Moreover, MemGPT [25] leverages cache-like architectures to prioritize recent information. Similarly, SCM [32] proposes a Self-Controlled Memory framework that enhances LLMs' capability to maintain long-term memory through a memory stream and controller mechanism. However, these approaches face significant limitations in handling diverse real-world tasks. While they can provide basic memory functionality, their operations are typically constrained by predefined structures and fixed workflows. These constraints stem from their reliance on rigid operational

(a)传统记忆系统。 **(b)** 我们提出的代理式记忆。

图1: 传统记忆系统需要在工作流中预定义记忆访问模式，这限制了它们对不同场景的适应性。相比之下，我们的A-MEM通过支持动态记忆操作增强了大语言模型代理的灵活性。

无法建立创新性连接或随着知识演变开发新的组织模式。这种僵化的结构，加上固定的代理工作流程，严重限制了这些系统泛化到新环境的能力，以及在长期交互中保持有效性的能力。随着LLM代理处理更复杂、开放式的任务，灵活的知识组织和持续适应变得至关重要，因此，如何设计一个灵活且通用的记忆系统以支持*LLM*代理的长期交互仍然是一个关键挑战。

在本文中，我们介绍了一种新型代理式记忆系统，命名为 A-MEM，该系统用于大语言模型代理，能够实现动态记忆结构化，而无需依赖静态、预定的记忆操作。我们的方法借鉴了卡片盒方法 [15, 1], 的灵感——这是一种复杂的知识管理系统，通过原子笔记和灵活的链接机制创建相互连接的信息网络。我们的系统引入了能动式记忆架构，为大语言模型代理提供自主和灵活的记忆管理能力。对于每个新的记忆，我们构建全面的笔记，该笔记整合了多种表示形式：包括结构化文本属性（包括多个属性和用于相似性匹配的嵌入向量）。然后 A-MEM 分析历史记忆库，基于语义相似性和共享属性建立有意义的连接。这种集成过程不仅创建了新的链接，还能够在新记忆被纳入时实现动态演化，它们可以触发现有记忆的上下文表示更新，使整个记忆系统能够随着时间的推移不断精炼和深化其理解。贡献总结如下：

• 我们提出了A-MEM，一个为LLM代理设计的代理式记忆系统，它能够实现上下文描述的自主生成、记忆连接的动态建立，以及基于新经历对现有记忆的智能进化。该系统为LLM代理赋予了长期交互能力，而无需预先设定记忆操作。

• 我们设计了一种代理式记忆更新机制，新记忆会自动触发两个关键操作：链接生成和记忆进化。链接生成通过识别共享属性和相似的上下文描述，自动建立记忆之间的连接。记忆进化使现有记忆能够随着新经历的解析而动态适应，从而产生更高阶的模式和属性。

• 我们使用长期对话数据集对系统进行了全面评估，通过六种不同的评估指标比较了六种基础模型的性能，展示了显著的改进。此外，我们还提供了T-SNE可视化，以说明我们的代理式记忆系统的结构化组织。

2 相关工作

2.1 LLM代理的记忆

以往关于大语言模型智能体记忆系统的相关研究探索了多种记忆管理和利用的机制 [23, 21, 8, 39]。部分方法实现了交互存储，通过密集检索模型 [39] 或读写记忆结构 [24] 维持全面的历史记录。此外，MemGPT [25] 利用缓存式架构来优先处理近期信息。类似地，SCM [32] 提出了一种自控式记忆框架，通过记忆流和控制器机制增强大语言模型维持长期记忆的能力。然而，这些方法在处理多样化的现实任务时面临显著局限。虽然它们可以提供基本的记忆功能，但它们的操作通常受限于预定义的结构和固定的流程。这些限制源于它们对刚性操作模式的依赖

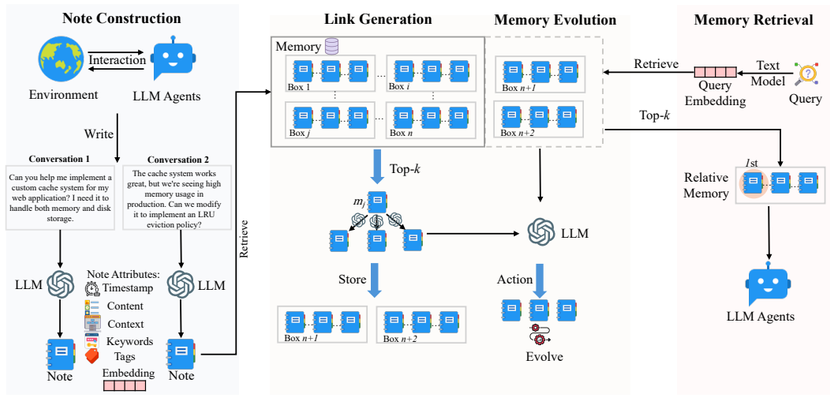

Figure 2: Our A-MEM architecture comprises three integral parts in memory storage. During note construction, the system processes new interaction memories and stores them as notes with multiple attributes. The link generation process first retrieves the most relevant historical memories and then employs an LLM to determine whether connections should be established between them. The concept of a 'box' describes that related memories become interconnected through their similar contextual descriptions, analogous to the Zettelkasten method. However, our approach allows individual memories to exist simultaneously within multiple different boxes. During the memory retrieval stage, we extract query embeddings using a text encoding model and search the memory database for relevant matches. When related memory is retrieved, similar memories that are linked within the same box are also automatically accessed.

patterns, particularly in memory writing and retrieval processes. Such inflexibility leads to poor generalization in new environments and limited effectiveness in long-term interactions. Therefore, designing a flexible and universal memory system that supports agents' long-term interactions remains a crucial challenge.

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has emerged as a powerful approach to enhance LLMs by incorporating external knowledge sources [18, 6, 10]. The standard RAG [37, 34] process involves indexing documents into chunks, retrieving relevant chunks based on semantic similarity, and augmenting the LLM's prompt with this retrieved context for generation. Advanced RAG systems [20, 12] have evolved to include sophisticated pre-retrieval and post-retrieval optimizations. Building upon these foundations, recent researches has introduced agentic RAG systems that demonstrate more autonomous and adaptive behaviors in the retrieval process. These systems can dynamically determine when and what to retrieve [4, 14], generate hypothetical responses to guide retrieval, and iteratively refine their search strategies based on intermediate results [31, 29].

However, while agentic RAG approaches demonstrate agency in the retrieval phase by autonomously deciding when and what to retrieve [4, 14, 38], our agentic memory system exhibits agency at a more fundamental level through the autonomous evolution of its memory structure. Inspired by the Zettelkasten method, our system allows memories to actively generate their own contextual descriptions, form meaningful connections with related memories, and evolve both their content and relationships as new experiences emerge. This fundamental distinction in agency between retrieval versus storage and evolution distinguishes our approach from agentic RAG systems, which maintain static knowledge bases despite their sophisticated retrieval mechanisms.

3 Methodolodgy

Our proposed agentic memory system draws inspiration from the Zettelkasten method, implementing a dynamic and self-evolving memory system that enables LLM agents to maintain long-term memory without predetermined operations. The system's design emphasizes atomic note-taking, flexible linking mechanisms, and continuous evolution of knowledge structures.

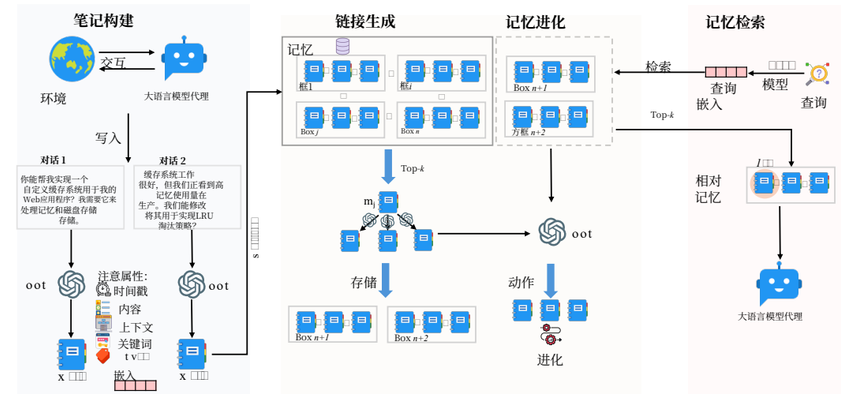

图2：我们的A-MEM架构在内存存储中包含三个核心部分。在笔记构建过程中，系统处理新的交互记忆，并将它们作为具有多种属性的笔记进行存储。链接生成过程首先检索最相关的历史记忆，然后使用一个大语言模型来确定它们之间是否应该建立连接。'盒子'的概念描述了相关记忆通过相似的上下文描述相互连接，类似于卡片盒方法。然而，我们的方法允许单个记忆同时存在于多个不同的盒子中。在记忆检索阶段，我们使用文本编码模型提取查询嵌入，并在记忆数据库中搜索相关匹配。当检索到相关记忆时，同一盒子内相互链接的相似记忆也会被自动访问。

，尤其是在记忆写入和检索过程中。这种不灵活性导致在新环境中泛化能力差，长期交互效果有限。因此，设计一个灵活且通用的记忆系统，支持智能体的长期交互，仍然是一个关键挑战。

2.2 基于检索的生成

检索增强生成 (RAG) 已成为一种强大的方法，通过整合外部知识源来增强大语言模型 [18, 6, 10]。标准 RAG [37, 34] 流程涉及将文档索引成块，根据语义相似性检索相关块，并将检索到的上下文用于增强大语言模型的提示以进行生成。高级 RAG 系统 [20,12] 已进化出复杂的预检索和后检索优化。在上述基础上，近期研究引入了具有自主性和适应性的智能 RAG 系统，在检索过程中表现出更智能的行为。这些系统可以动态确定何时以及检索什么内容 [4, 14], 生成假设性响应以指导检索，并根据中间结果迭代优化其搜索策略 [31, 29]。

然而，虽然代理式检索增强生成（agentic RAG）方法在检索阶段通过自主决定何时以及检索什么来展现能动性，但我们的代理式记忆系统通过其记忆结构的自主进化，在更根本的层面上展现能动性。受卡片盒方法（Zettelkasten method）的启发，我们的系统允许记忆主动生成自己的上下文描述，与相关记忆建立有意义的联系，并在新经历出现时进化其内容与关系。这种检索与存储及进化之间的能动性根本区别，将我们的方法与代理式RAG系统区分开来，后者尽管拥有复杂的检索机制，但仍然维持静态的知识库。

3 方法论

我们提出的代理式记忆系统借鉴了卡片盒方法，实现了一个动态且自我进化的记忆系统，使大语言模型代理能够在没有预定操作的情况下保持长期记忆。该系统的设计强调原子笔记记录、灵活的链接机制以及知识结构的持续进化。

3.1 Note Construction

Building upon the Zettelkasten method's principles of atomic note-taking and flexible organization, we introduce an LLM-driven approach to memory note construction. When an agent interacts with its environment, we construct structured memory notes that capture both explicit information and LLM-generated contextual understanding. Each memory note m_i in our collection $\mathcal{M} = \{m_1, m_2, ..., m_N\}$ is represented as:

$$m_i = \{c_i, t_i, K_i, G_i, X_i, e_i, L_i\} \tag{1}$$

where c_i represents the original interaction content, t_i is the timestamp of the interaction, K_i denotes LLM-generated keywords that capture key concepts, G_i contains LLM-generated tags for categorization, X_i represents the LLM-generated contextual description that provides rich semantic understanding, and L_i maintains the set of linked memories that share semantic relationships. To enrich each memory note with meaningful context beyond its basic content and timestamp, we leverage an LLM to analyze the interaction and generate these semantic components. The note construction process involves prompting the LLM with carefully designed templates P_{s1}:

$$K_i, G_i, X_i \leftarrow \text{LLM}(c_i \,\|t_i\, \|P_{s1}) \tag{2}$$

Following the Zettelkasten principle of atomicity, each note captures a single, self-contained unit of knowledge. To enable efficient retrieval and linking, we compute a dense vector representation via a text encoder [27] that encapsulates all textual components of the note:

$$e_i = f_{\text{enc}}[\ \text{concat}(c_i, K_i, G_i, X_i)\ ] \tag{3}$$

By using LLMs to generate enriched components, we enable autonomous extraction of implicit knowledge from raw interactions. The multi-faceted note structure (K_i, G_i, X_i) creates rich representations that capture different aspects of the memory, facilitating nuanced organization and retrieval. Additionally, the combination of LLM-generated semantic components with dense vector representations provides both context and computationally efficient similarity matching.

3.2 Link Generation

Our system implements an autonomous link generation mechanism that enables new memory notes to form meaningful connections without predefined rules. When the constrctd memory note m_n is added to the system, we first leverage its semantic embedding for similarity-based retrieval. For each existing memory note $m_j \in \mathcal{M}$, we compute a similarity score:

$$s_{n,j} = \frac{e_n \cdot e_j}{|e_n||e_j|} \tag{4}$$

The system then identifies the top-k most relevant memories:

$$\mathcal{M}^n_{\text{near}} = \{m_j | \ \text{rank}(s_{n,j}) \leq k, m_j \in \mathcal{M}\} \tag{5}$$

Based on these candidate nearest memories, we prompt the LLM to analyze potential connections based on their potential common attributes. Formally, the link set of memory m_n update like:

$$L_i \leftarrow \text{LLM}(m_n \,\|\mathcal{M}^n_{\text{near}}\, \|P_{s2}) \tag{6}$$

Each generated link l_i is structured as: $L_i = \{m_i, ..., m_k\}$. By using embedding-based retrieval as an initial filter, we enable efficient scalability while maintaining semantic relevance. A-MEM can quickly identify potential connections even in large memory collections without exhaustive comparison. More importantly, the LLM-driven analysis allows for nuanced understanding of relationships that goes beyond simple similarity metrics. The language model can identify subtle patterns, causal relationships, and conceptual connections that might not be apparent from embedding similarity alone. We implements the Zettelkasten principle of flexible linking while leveraging modern language models. The resulting network emerges organically from memory content and context, enabling natural knowledge organization.

3.3 Memory Evolution

After creating links for the new memory, A-MEM evolves the retrieved memories based on their textual information and relationships with the new memory. For each memory m_j in the nearest

3.1 笔记构建

基于卡片盒方法原子笔记记录和灵活组织的原则，我们引入了一种由大语言模型驱动的记忆笔记构建方法。当代理与环境交互时，我们构建结构化的记忆笔记，这些笔记既捕获显式信息，也捕获大语言模型生成的上下文理解。我们集合中的每个记忆笔记 m_i $\mathcal{M} = \{m_1, m_2, ..., m_N\}$ 都表示为：

$$m_i = \{c_i, t_i, K_i, G_i, X_i, e_i, L_i\} \tag{1}$$

其中 c_i 代表原始交互内容， t_i 是交互的时间戳， K_i 表示大语言模型生成的用于捕捉关键概念的关键词， G_i 包含大语言模型生成的用于分类的标签， X_i 代表大语言模型生成的用于提供丰富语义理解的上下文描述，而 L_i 维持了一组具有语义关联的链接记忆。为了在记忆笔记的基本内容和时间戳之外为其添加有意义的上下文，我们利用大语言模型分析交互并生成这些语义组件。笔记构建过程涉及使用精心设计的模板 P_{s1} 提示大语言模型：

$$K_i, G_i, X_i \leftarrow \text{LLM}(c_i \,\|t_i\, \|P_{s1}) \tag{2}$$

遵循原子性原则的 Zettelkasten 方法，每条笔记捕获一个单一、自包含的知识单元。为了实现高效的检索和链接，我们通过文本编码器计算密集向量表示 [27] 该向量封装了笔记的所有文本组件：

$$e_i = f_{\text{enc}}[\ \text{concat}(c_i, K_i, G_i, X_i)\ ] \tag{3}$$

通过使用大语言模型生成丰富的组件，我们实现了从原始交互中自主提取隐含知识。多方面的笔记结构 (K_i, G_i, X_i) 创建了捕捉记忆不同方面的丰富表示，促进了细致的组织和检索。此外，大语言模型生成的语义组件与密集向量表示的结合提供了上下文和计算上高效的相似度匹配。

3.2 链接生成

我们的系统实现了一种自主的链接生成机制，使新的记忆笔记能够在没有预定义规则的情况下形成有意义的连接。当构建好的记忆笔记 m_n 被添加到系统中时，我们首先利用其语义嵌入进行基于相似度的检索。对于每个现有的记忆笔记 $m_j \in \mathcal{M}$，我们计算一个相似度分数：

$$s_{n,j} = \frac{e_n \cdot e_j}{|e_n||e_j|} \tag{4}$$

系统随后识别出最-k 个最相关的记忆：

$$\mathcal{M}^n_{\text{near}} = \{m_j | \ \text{rank}(s_{n,j}) \leq k, m_j \in \mathcal{M}\} \tag{5}$$

基于这些候选最近记忆，我们提示大语言模型分析基于它们潜在共同属性的可能连接。形式上，记忆 m_n 的链接集更新如下：

$$L_i \leftarrow \text{LLM}(m_n \,\|\mathcal{M}^n_{\text{near}}\, \|P_{s2}) \tag{6}$$

每个生成的链接 l_i 的结构为： $L_i = \{m_i, ..., m_k\}$。通过使用基于嵌入的检索作为初始过滤器，我们实现了高效的扩展性，同时保持了语义相关性。A-MEM可以快速识别大型记忆集合中的潜在连接，而无需进行穷举比较。更重要的是，由大语言模型驱动的分析允许对关系进行细致的理解，这超出了简单的相似度指标。语言模型可以识别微妙的模式、因果关系和概念连接，这些可能仅从嵌入相似度中不明显。我们实现了灵活链接的Zettelkasten原则，同时利用现代语言模型。由此产生的网络从记忆内容和上下文中自然涌现，实现了自然的知识组织。

3.3 记忆进化

在为新的记忆创建链接后，A-MEM 会根据其文本信息以及与新记忆的关系来进化检索到的记忆。对于最近邻集合 m_j 中的每条记忆m_j，系统会确定是否更新其上下文、关键词和标签。

neighbor set $\mathcal{M}^n_{\text{near}}$, the system determines whether to update its context, keywords, and tags. This evolution process can be formally expressed as:

$$m_j^* \leftarrow \text{LLM}(m_n \,\|\mathcal{M}^n_{\text{near}} \setminus m_j \,\|m_j \,\|P_{s3}) \tag{7}$$

The evolved memory m_j^* then replaces the original memory m_j in the memory set $\mathcal{M}$. This evolutionary approach enables continuous updates and new connections, mimicking human learning processes. As the system processes more memories over time, it develops increasingly sophisticated knowledge structures, discovering higher-order patterns and concepts across multiple memories. This creates a foundation for autonomous memory learning where knowledge organization becomes progressively richer through the ongoing interaction between new experiences and existing memories.

3.4 Retrieve Relative Memory

In each interaction, our A-MEM performs context-aware memory retrieval to provide the agent with relevant historical information. Given a query text q from the current interaction, we first compute its dense vector representation using the same text encoder used for memory notes:

$$e_q = f_{\text{enc}}(q) \tag{8}$$

The system then computes similarity scores between the query embedding and all existing memory notes in $\mathcal{M}$ using cosine similarity:

$$s_{q,i} = \frac{e_q \cdot e_i}{|e_q||e_i|}, \text{where } e_i \in m_i,\ \forall m_i \in \mathcal{M} \tag{9}$$

Then we retrieve the k most relevant memories from the historical memory storage to construct a contextually appropriate prompt.

$$\mathcal{M}_{\text{retrieved}} = \{m_i | \text{rank}(s_{q,i}) \leq k, m_i \in \mathcal{M}\} \tag{10}$$

These retrieved memories provide relevant historical context that helps the agent better understand and respond to the current interaction. The retrieved context enriches the agent's reasoning process by connecting the current interaction with related past experiences stored in the memory system.

4 Experiment

4.1 Dataset and Evaluation

To evaluate the effectiveness of instruction-aware recommendation in long-term conversations, we utilize the LoCoMo dataset [22], which contains significantly longer dialogues compared to existing conversational datasets [36, 13]. While previous datasets contain dialogues with around 1K tokens over 4-5 sessions, LoCoMo features much longer conversations averaging 9K tokens spanning up to 35 sessions, making it particularly suitable for evaluating models' ability to handle long-range dependencies and maintain consistency over extended conversations. The LoCoMo dataset comprises diverse question types designed to comprehensively evaluate different aspects of model understanding: (1) single-hop questions answerable from a single session; (2) multi-hop questions requiring information synthesis across sessions; (3) temporal reasoning questions testing understanding of time-related information; (4) open-domain knowledge questions requiring integration of conversation context with external knowledge; and (5) adversarial questions assessing models' ability to identify unanswerable queries. In total, LoCoMo contains 7,512 question-answer pairs across these categories. Besides, we use a new dataset, named DialSim [16], to evaluate the effectiveness of our memory system. It is question-answering dataset derived from long-term multi-party dialogues. The dataset is derived from popular TV shows (Friends, The Big Bang Theory, and The Office), covering 1,300 sessions spanning five years, containing approximately 350,000 tokens, and including more than 1,000 questions per session from refined fan quiz website questions and complex questions generated from temporal knowledge graphs.

For comparison baselines, we compare to **LoCoMo** [22], **ReadAgent** [17], **MemoryBank** [39] and **MemGPT** [25]. The detailed introduction of baselines can be found in Appendix A.1 For evaluation, we employ two primary metrics: the F1 score to assess answer accuracy by balancing precision and recall, and BLEU-1 [26] to evaluate generated response quality by measuring word overlap

该进化过程可以用以下公式正式表示： $\mathcal{M}^n$ 近

$$m_j^* \leftarrow \text{LLM}(m_n \,\|\mathcal{M}^n_{\text{near}} \setminus m_j \,\|m_j \,\|P_{s3}) \tag{7}$$

进化后的记忆 m_j^* 会替换记忆集合 $\mathcal{M}$ 中的原始记忆 m_j 。这种进化方法支持持续更新和建立新连接，模拟人类学习过程。随着系统处理更多记忆，它会发展出越来越复杂的知识结构，在多个记忆中发现高阶模式和概念。这为自主记忆学习奠定了基础，通过新经历与现有记忆之间持续的交互，知识组织变得越来越丰富。

3.4 检索相对记忆

在每次交互中，我们的 A-MEM 执行上下文感知的记忆检索，为智能体提供相关的历史信息。给定当前交互中的查询文本 q ，我们首先使用用于记忆笔记的相同文本编码器计算其密集向量表示：

$$e_q = f_{\text{enc}}(q) \tag{8}$$

系统接着使用余弦相似度计算查询嵌入与 $\mathcal{M}$ 中所有现有记忆笔记之间的相似度分数：

$$s_{q,i} = \frac{e_q \cdot e_i}{|e_q||e_i|}, \text{where } e_i \in m_i,\ \forall m_i \in \mathcal{M} \tag{9}$$

然后我们从历史内存存储中检索 k 个最相关的记忆，以构建上下文合适的提示。

$$\mathcal{M}_{\text{retrieved}} = \{m_i | \text{rank}(s_{q,i}) \leq k, m_i \in \mathcal{M}\} \tag{10}$$

这些检索到的记忆提供了相关的历史上下文，帮助智能体更好地理解和回应当前交互。检索到的上下文通过将当前交互与记忆系统中存储的相关过去经历连接起来，丰富了智能体的推理过程。

4 实验

4.1 数据集与评估

为了评估指令感知推荐在长期对话中的有效性，我们利用了LoCoMo数据集 [22], 该数据集包含的对话比现有对话数据集显著更长 [36, 13]。而之前的数据集包含约1K个token的4-5个会话的对话，LoCoMo则具有平均9K个token、长达35个会话的更长对话，这使得它特别适合评估模型处理长距离依赖关系和在整个对话中保持一致性的能力。LoCoMo数据集包含多种问题类型，旨在全面评估模型理解的各个方面：(1) 单跳问题，可从单个会话中回答；(2) 多跳问题，需要跨会话综合信息；(3) 时序推理问题，测试对时间相关信息的理解；(4) 开放域知识问题，需要将对话上下文与外部知识相结合；(5) 对抗性问题，评估模型识别无解查询的能力。总共有7,512个问答对属于这些类别。此外，我们使用了一个新数据集，名为DialSim [16], 来评估我们记忆系统的有效性。它是一个问答数据集，源自长期多方对话。该数据集来自热门电视剧（《老友记》、《生活大爆炸》和《办公室》），涵盖1,300个会话，跨越五年，包含约350,000个token，并且每个会话包含超过1,000个问题，这些问题来自经过精炼的粉丝问答网站问题和从时序知识图谱生成的复杂问题。

为了进行比较基线，我们将其与 **LoCoMo** [22], **ReadAgent** [17], **MemoryBank** [39] 以及**MemGPT** [25]进行比较。基线的详细介绍可以在附录A.1中找到。在评估方面，我们采用两个主要指标：F1分数用于通过平衡精确率和召回率来评估答案的准确性，以及BLEU-1 [26] 用于通过测量词重叠来评估生成响应的质量

Table 1: Experimental results on LoCoMo dataset of QA tasks across five categories (Multi Hop, Temporal, Open Domain, Single Hop, and Adversial) using different methods. Results are reported in F1 and BLEU-1 (%) scores. The best performance is marked in bold, and our proposed method A-MEM (highlighted in gray) demonstrates competitive performance across six foundation language models.

Model		Method	Category										Average		
			Multi Hop		Temporal		Open Domain		Single Hop		Adversial		Ranking		Token
			F1	BLEU	F1	BLEU	F1	BLEU	F1	BLEU	F1	BLEU	F1	BLEU	Length
GPT	4o-mini	LoCoMo	25.02	19.75	18.41	14.77	12.04	11.16	40.36	29.05	**69.23**	**68.75**	2.4	2.4	16,910
		ReadAgent	9.15	6.48	12.60	8.87	5.31	5.12	9.67	7.66	9.81	9.02	4.2	4.2	643
		MemoryBank	5.00	4.77	9.68	6.99	5.56	5.94	6.61	5.16	7.36	6.48	4.8	4.8	432
		MemGPT	26.65	17.72	25.52	19.44	9.15	7.44	41.04	34.34	43.29	42.73	2.4	2.4	16,977
		A-Mem	**27.02**	**20.09**	**45.85**	**36.67**	**12.14**	**12.00**	**44.65**	**37.06**	50.03	49.47	**1.2**	**1.2**	2,520
	4o	LoCoMo	28.00	18.47	9.09	5.78	16.47	14.80	**61.56**	**54.19**	**52.61**	**51.13**	2.0	2.0	16,910
		ReadAgent	14.61	9.95	4.16	3.19	8.84	8.37	12.46	10.29	6.81	6.13	4.0	4.0	805
		MemoryBank	6.49	4.69	2.47	2.43	6.43	5.30	8.28	7.10	4.42	3.67	5.0	5.0	569
		MemGPT	30.36	22.83	17.29	13.18	12.24	11.87	60.16	53.35	34.96	34.25	2.4	2.4	16,987
		A-Mem	**32.86**	**23.76**	**39.41**	**31.23**	**17.10**	**15.84**	48.43	42.97	36.35	35.53	**1.6**	**1.6**	1,216
Qwen2.5	1.5b	LoCoMo	9.05	6.55	4.25	4.04	9.91	8.50	11.15	8.67	40.38	40.23	3.4	3.4	16,910
		ReadAgent	6.61	4.93	2.55	2.51	5.31	12.24	10.13	7.54	5.42	27.32	4.6	4.6	752
		MemoryBank	11.14	8.25	4.46	2.87	8.05	6.21	13.42	11.01	36.76	34.00	2.6	2.6	284
		MemGPT	10.44	7.61	4.21	3.89	13.42	11.64	9.56	7.34	31.51	28.90	3.4	3.4	16,953
		A-Mem	**18.23**	**11.94**	**24.32**	**19.74**	**16.48**	**14.31**	**23.63**	**19.23**	**46.00**	**43.26**	**1.0**	**1.0**	1,300
	3b	LoCoMo	4.61	4.29	3.11	2.71	4.55	5.97	7.03	5.69	16.95	14.81	3.2	3.2	16,910
		ReadAgent	2.47	1.78	3.01	3.01	5.57	5.22	3.25	2.51	15.78	14.01	4.2	4.2	776
		MemoryBank	3.60	3.39	1.72	1.97	6.63	6.58	4.11	3.32	13.07	10.30	4.2	4.2	298
		MemGPT	5.07	4.31	2.94	2.95	7.04	7.10	7.26	5.52	14.47	12.39	2.4	2.4	16,961
		A-Mem	**12.57**	**9.01**	**27.59**	**25.07**	**7.12**	**7.28**	**17.23**	**13.12**	**27.91**	**25.15**	**1.0**	**1.0**	1,137
Llama 3.2	1b	LoCoMo	11.25	9.18	7.38	6.82	11.90	10.38	12.86	10.50	51.89	48.27	3.4	3.4	16,910
		ReadAgent	5.96	5.12	1.93	2.30	12.46	11.17	7.75	6.03	44.64	40.15	4.6	4.6	665
		MemoryBank	13.18	10.03	7.61	6.27	15.78	12.94	17.30	14.03	52.61	47.53	2.0	2.0	274
		MemGPT	9.19	6.96	4.02	4.79	11.14	8.24	10.16	7.68	49.75	45.11	4.0	4.0	16,950
		A-Mem	**19.06**	**11.71**	**17.80**	**10.28**	**17.55**	**14.67**	**28.51**	**24.13**	**58.81**	**54.28**	**1.0**	**1.0**	1,376
	3b	LoCoMo	6.88	5.77	4.37	4.40	10.65	9.29	8.37	6.93	30.25	28.46	2.8	2.8	16,910
		ReadAgent	2.47	1.78	3.01	3.01	5.57	5.22	3.25	2.51	15.78	14.01	4.2	4.2	461
		MemoryBank	6.19	4.47	3.49	3.13	4.07	4.57	7.61	6.03	18.65	17.05	3.2	3.2	263
		MemGPT	5.32	3.99	2.68	2.72	5.64	5.54	4.32	3.51	21.45	19.37	3.8	3.8	16,956
		A-Mem	**17.44**	**11.74**	**26.38**	**19.50**	**12.53**	**11.83**	**28.14**	**23.87**	**42.04**	**40.60**	**1.0**	**1.0**	1,126

with ground truth responses. Also, we report the average token length for answering one question. Besides reporting experiment results with four additional metrics (ROUGE-L, ROUGE-2, METEOR, and SBERT Similarity), we also present experimental outcomes using different foundation models including DeepSeek-R1-32B [11], Claude 3.0 Haiku [2], and Claude 3.5 Haiku [3] in Appendix A.3.

4.2 Implementation Details

For all baselines and our proposed method, we maintain consistency by employing identical system prompts as detailed in Appendix B. The deployment of Qwen-1.5B/3B and Llama 3.2 1B/3B models is accomplished through local instantiation using Ollama [1], with LiteLLM [2] managing structured output generation. For GPT models, we utilize the official structured output API. In our memory retrieval process, we primarily employ k=10 for top-k memory selection to maintain computational efficiency, while adjusting this parameter for specific categories to optimize performance. The detailed configurations of k can be found in Appendix A.5. For text embedding, we implement the all-minilm-l6-v2 model across all experiments.

4.3 Empricial Results

Performance Analysis. In our empirical evaluation, we compared A-MEM with four competitive baselines including LoCoMo [22], ReadAgent [17], MemoryBank [39], and MemGPT [25] on the LoCoMo dataset. For non-GPT foundation models, our A-MEM consistently outperforms all baselines across different categories, demonstrating the effectiveness of our agentic memory approach. For GPT-based models, while LoCoMo and MemGPT show strong performance in certain categories like Open Domain and Adversial tasks due to their robust pre-trained knowledge in simple fact retrieval, our A-MEM demonstrates superior performance in Multi-Hop tasks achieves at least two times better performance that require complex reasoning chains. In addition to experiments on the LoCoMo dataset, we also compare our method on the DialSim dataset against LoCoMo and MemGPT. A-MEM consistently outperforms all baselines across evaluation metrics, achieving an F1

[1] https://github.com/ollama/ollama
[2] https://github.com/BerriAI/litellm

表1： 在LoCoMo数据集上针对问答任务（包含多跳、时序、开放域、单跳和对抗五个类别）使用不同方法进行的实验结果。结果以F1和BLEU-1 (%) 分数报告。最佳性能以粗体标出，我们提出的方法A-MEM（以灰色突出显示）在六种基础语言模型上均表现出具有竞争力的性能。

模型		方法	分类										平均		
			多跳		时序		开放域		单跳		对抗		排序		标记
			F1	BLEU	F1	BLEU	F1	BLEU	F1	BLEU	F1	BLEU	F1分数	BLEU	长度
GPT	4o-mini	LOCOMO	25.02	19.75	18.41	14.77	12.04	11.16	40.36	29.05	**69.23**	**68.75**	2.4	2.4	16,910
		READAGENT	9.15	6.48	12.60	8.87	5.31	5.12	9.67	7.66	9.81	9.02	4.2	4.2	643
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		A-MEM	**27.02**	**20.09**	**45.85**	**36.67**	**12.14**	**12.00**	**44.65**	**37.06**	50.03	49.47	**1.2**	**1.2**	2,520
	4o	LOCOMO	28.00	18.47	9.09	5.78	16.47	14.80	**61.56**	**54.19**	**52.61**	**51.13**	2.0	2.0	16,910
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		MEMORYBANK	6.49	4.69	2.47	2.43	6.43	5.30	8.28	7.10	4.42	3.67	5.0	5.0	569
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		A-MEM	**32.86**	**23.76**	**39.41**	**31.23**	**17.10**	**15.84**	48.43	42.97	36.35	35.53	**1.6**	**1.6**	1,216
Qwen2.5	1.5b	LOCOMO	9.05	6.55	4.25	4.04	9.91	8.50	11.15	8.67	40.38	40.23	3.4	3.4	16,910
		READAGENT	6.61	4.93	2.55	2.51	5.31	12.24	10.13	7.54	5.42	27.32	4.6	4.6	752
		MEMORYBANK	11.14	8.25	4.46	2.87	8.05	6.21	13.42	11.01	36.76	34.00	2.6	2.6	284
		MEMGPT	10.44	7.61	4.21	3.89	13.42	11.64	9.56	7.34	31.51	28.90	3.4	3.4	16,953
		A-MEM	**18.23**	**11.94**	**24.32**	**19.74**	**16.48**	**14.31**	**23.63**	**19.23**	**46.00**	**43.26**	**1.0**	**1.0**	1,300
	3b	LOCOMO	4.61	4.29	3.11	2.71	4.55	5.97	7.03	5.69	16.95	14.81	3.2	3.2	16,910
		读取代理	2.47	1.78	3.01	3.01	5.57	5.22	3.25	2.51	15.78	14.01	4.2	4.2	776
		记忆库	3.60	3.39	1.72	1.97	6.63	6.58	4.11	3.32	13.07	10.30	4.2	4.2	298
		MEMGPT	5.07	4.31	2.94	2.95	7.04	7.10	7.26	5.52	14.47	12.39	2.4	2.4	16,961
		A-MEM	**12.57**	**9.01**	**27.59**	**25.07**	**7.12**	**7.28**	**17.23**	**13.12**	**27.91**	**25.15**	**1.0**	**1.0**	1,137
Llama 3.2	1b	LOCOMO	11.25	9.18	7.38	6.82	11.90	10.38	12.86	10.50	51.89	48.27	3.4	3.4	16,910
		读取代理	5.96	5.12	1.93	2.30	12.46	11.17	7.75	6.03	44.64	40.15	4.6	4.6	665
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		MEMGPT	9.19	6.96	4.02	4.79	11.14	8.24	10.16	7.68	49.75	45.11	4.0	4.0	16,950
		A-MEM	**19.06**	**11.71**	**17.80**	**10.28**	**17.55**	**14.67**	**28.51**	**24.13**	**58.81**	**54.28**	**1.0**	**1.0**	1,376
	3b	LOCOMO	6.88	5.77	4.37	4.40	10.65	9.29	8.37	6.93	30.25	28.46	2.8	2.8	16,910
		读取代理	2.47	1.78	3.01	3.01	5.57	5.22	3.25	2.51	15.78	14.01	4.2	4.2	461
		MEMORYBANK	6.19	4.47	3.49	3.13	4.07	4.57	7.61	6.03	18.65	17.05	3.2	3.2	263
		MEMGPT	5.32	3.99	2.68	2.72	5.64	5.54	4.32	3.51	21.45	19.37	3.8	3.8	16,956
		A-MEM	**17.44**	**11.74**	**26.38**	**19.50**	**12.53**	**11.83**	**28.14**	**23.87**	**42.04**	**40.60**	**1.0**	**1.0**	1,126

与真实响应进行比较。此外，我们还报告了回答一个问题所需的平均token长度。除了报告使用四个附加指标（ROUGE-L、ROUGE-2、METEOR和SBERT相似度）的实验结果外，我们还展示了使用不同基础模型（包括DeepSeek-R1-32B [11], Claude 3.0Haiku [2], 和Claude 3.5 Haiku [3]）的实验结果A.3。

4.2 实现细节

对于所有基线模型和我们的提出的方法，我们通过采用附录B中详细说明的相同系统提示来保持一致性1，通过Ollama [2] 本地实例化部署Qwen-1.5B/3B和Llama 3.2 1B/3B模型1，使用LiteLLM [2] 管理结构化输出生成。对于GPT模型，我们使用官方结构化输出API。在我们的记忆检索过程中，我们主要采用 k=10 用于顶部k 记忆选择以保持计算效率，同时针对特定类别调整此参数以优化性能。的详细配置可以在附录 k A.5中找到。对于文本嵌入，我们在所有实验中都实现了all-minilm-l6-v2模型。

4.3 实验结果

性能分析。 在我们的实证评估中，我们比较了 A-MEM 与四个竞争性基线，包括 LoCoMo [22], ReadAgent [17], MemoryBank [39], 和 MemGPT [25] 在 LoCoMo 数据集上的表现。对于非 GPT 基础模型，我们的 A-MEM 在不同类别中始终优于所有基线，证明了我们代理式记忆方法的有效性。对于基于 GPT 的模型，虽然 LoCoMo 和 MemGPT 在开放域和对抗任务等某些类别中表现出色，这得益于它们在简单事实检索方面强大的预训练知识，但我们的 A-MEM 在多跳任务中表现更优，其性能至少比需要复杂推理链的任务高两倍。除了在 LoCoMo 数据集上的实验，我们还将在 DialSim 数据集上与 LoCoMo 和 MemGPT 进行方法比较。A-MEM 在所有评估指标中始终优于所有基线，F1 分数为 3.45（比 LoCoMo 的 2.55 高 35%，比 MemGPT 的 1.18 高 192%）。

[1] https://github.com/ollama/ollama [2]
https://github.com/BerriAI/litellm

Table 2: Comparison of different memory mechanisms across multiple evaluation metrics on DialSim [16]. Higher scores indicate better performance, with A-MEM showing superior results across all metrics.

Method	F1	BLEU-1	ROUGE-L	ROUGE-2	METEOR	SBERT Similarity
LoCoMo	2.55	3.13	2.75	0.90	1.64	15.76
MemGPT	1.18	1.07	0.96	0.42	0.95	8.54
A-MEM	**3.45**	**3.37**	**3.54**	**3.60**	**2.05**	**19.51**

Table 3: An ablation study was conducted to evaluate our proposed method against the GPT-4o-mini base model. The notation 'w/o' indicates experiments where specific modules were removed. The abbreviations LG and ME denote the link generation module and memory evolution module, respectively.

Method	Category									
	Multi Hop		Temporal		Open Domain		Single Hop		Adversial	
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
w/o LG & ME	9.65	7.09	24.55	19.48	7.77	6.70	13.28	10.30	15.32	18.02
w/o ME	21.35	15.13	31.24	27.31	10.13	10.85	39.17	34.70	44.16	45.33
A-MEM	**27.02**	**20.09**	**45.85**	**36.67**	**12.14**	**12.00**	**44.65**	**37.06**	**50.03**	**49.47**

score of 3.45 (a 35% improvement over LoCoMo's 2.55 and 192% higher than MemGPT's 1.18). The effectiveness of A-MEM stems from its novel agentic memory architecture that enables dynamic and structured memory management. Unlike traditional approaches that use static memory operations, our system creates interconnected memory networks through atomic notes with rich contextual descriptions, enabling more effective multi-hop reasoning. The system's ability to dynamically establish connections between memories based on shared attributes and continuously update existing memory descriptions with new contextual information allows it to better capture and utilize the relationships between different pieces of information.

Cost-Efficiency Analysis. A-MEM demonstrates significant computational and cost efficiency alongside strong performance. The system requires approximately 1,200 tokens per memory operation, achieving an 85-93% reduction in token usage compared to baseline methods (LoCoMo and MemGPT with 16,900 tokens) through our selective top-k retrieval mechanism. This substantial token reduction directly translates to lower operational costs, with each memory operation costing less than $0.0003 when using commercial API services—making large-scale deployments economically viable. Processing times average 5.4 seconds using GPT-4o-mini and only 1.1 seconds with locally-hosted Llama 3.2 1B on a single GPU. Despite requiring multiple LLM calls during memory processing, A-MEM maintains this cost-effective resource utilization while consistently outperforming baseline approaches across all foundation models tested, particularly doubling performance on complex multi-hop reasoning tasks. This balance of low computational cost and superior reasoning capability highlights A-MEM's practical advantage for deployment in the real world.

4.4 Ablation Study

To evaluate the effectiveness of the Link Generation (LG) and Memory Evolution (ME) modules, we conduct the ablation study by systematically removing key components of our model. When both LG and ME modules are removed, the system exhibits substantial performance degradation, particularly in Multi Hop reasoning and Open Domain tasks. The system with only LG active (w/o ME) shows intermediate performance levels, maintaining significantly better results than the version without both modules, which demonstrates the fundamental importance of link generation in establishing memory connections. Our full model, A-MEM, consistently achieves the best performance across all evaluation categories, with particularly strong results in complex reasoning tasks. These results reveal that while the link generation module serves as a critical foundation for memory organization, the memory evolution module provides essential refinements to the memory structure. The ablation study validates our architectural design choices and highlights the complementary nature of these two modules in creating an effective memory system.

4.5 Hyperparameter Analysis

We conducted extensive experiments to analyze the impact of the memory retrieval parameter k, which controls the number of relevant memories retrieved for each interaction. As shown in Figure 3, we evaluated performance across different k values (10, 20, 30, 40, 50) on five categories of tasks using GPT-4o-mini as our base model. The results reveal an interesting pattern: while increasing k generally leads to improved performance, this improvement gradually plateaus and sometimes slightly decreases at higher values. This trend is particularly evident in Multi Hop and Open Domain

表2： 在DialSim上，不同记忆机制在多个评估指标上的比较 [16]。分数越高表示性能越好，A-MEM在所有指标上均表现优异。

方法	F1	BLEU-1	ROUGE-L	ROUGE-2	METEOR	SBERT相似度
LoCoMo	2.55	3.13	2.75	0.90	1.64	15.76
MemGPT	1.18	1.07	0.96	0.42	0.95	8.54
A-MEM	**3.45**	**3.37**	**3.54**	**3.60**	**2.05**	**19.51**

表3： 为评估我们提出的方法相对于GPT-4o-mini基础模型的性能，进行了消融研究。符号'w/o'表示移除了特定模块的实验。缩写LG和ME分别表示链接生成模块和记忆进化模块。

方法	类别									
	多跳		时序		开放域		单跳		对抗	
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
w/o 链接生成 & 记忆进化	9.65	7.09	24.55	19.48	7.77	6.70	13.28	10.30	15.32	18.02
w/o 记忆进化	21.35	15.13	31.24	27.31	10.13	10.85	39.17	34.70	44.16	45.33
A-MEM	**27.02**	**20.09**	**45.85**	**36.67**	**12.14**	**12.00**	**44.65**	**37.06**	**50.03**	**49.47**

A-MEM 的有效性源于其新颖的代理式记忆架构，该架构能够实现动态和结构化的记忆管理。与使用静态记忆操作的 传统方法不同，我们的系统通过原子笔记和丰富的上下文描述创建了相互连接的记忆网络，从而实现更有效的多跳推理。该系统能够根据共享属性动态建立记忆之间的连接，并持续用新的上下文信息更新现有记忆描述，这使其能够更好地捕捉和利用不同信息片段之间的关系。

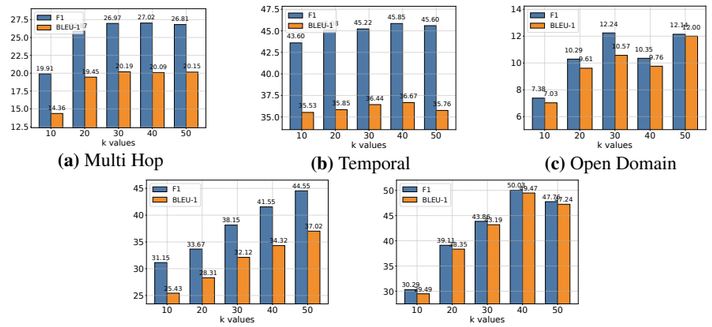
成本效益分析。 A-MEM在计算效率和成本方面表现出显著优势，同时保持优异的性能。该系统每项记忆操作需要约1,200个token，通过我们选择的top-k检索机制，与基线方法（LoCoMo和MemGPT使用16,900个token）相比，token使用量减少了85-93%。这种大幅度的token减少直接转化为更低的运营成本，在使用商业API服务时，每项记忆操作成本低于$0.0003，使得大规模部署在经济上可行。使用GPT-4o-mini处理时间平均为5.4秒，而在单个GPU上本地部署Llama 3.2 1B仅需1.1秒。尽管在记忆处理过程中需要多次调用LLM，A-MEM仍能保持这种具有成本效益的资源利用率，同时在所有测试的基础模型中始终优于基线方法，尤其是在复杂的多跳推理任务上性能提升了一倍。这种低计算成本与卓越推理能力的平衡突显了A-MEM在实际应用中的优势。

4.4 消融研究

为评估链接生成（LG）和记忆进化（ME）模块的有效性，我们通过系统地移除模型的关键组件来开展消融研究。当同时移除LG和ME模块时，系统表现出显著的性能下降，特别是在多跳推理和开放域任务中。仅激活LG模块（w/o ME）的系统显示中等性能水平，其结果明显优于同时移除这两个模块的版本，这证明了链接生成在建立记忆连接中的基础重要性。我们的完整模型A-MEM在所有评估类别中始终达到最佳性能，在复杂推理任务中尤其表现出色。这些结果表明，虽然链接生成模块是记忆组织的关键基础，但记忆进化模块为记忆结构提供了必要的改进。消融研究验证了我们的架构设计选择，并突出了这两个模块在构建有效记忆系统中的互补性。

4.5 超参数分析

我们进行了广泛的实验，以分析记忆检索参数k的影响，该参数控制每次交互检索的相关记忆数量。如图3所示，我们使用GPT-4o-mini作为基础模型，在五种任务类别上评估了不同k值（10、20、30、40、50）的性能。结果表明存在一个有趣的模式：虽然增加k通常会提高性能，但这种改进在高值时会逐渐趋于平稳，有时甚至会略微下降。这一趋势在多跳和开放域任务中尤为明显



(a) Multi Hop **(b)** Temporal **(c)** Open Domain

(d) Single Hop **(e)** Adversarial

Figure 3: Impact of memory retrieval parameter k across different task categories with GPT-4o-mini as the base model. While larger k values generally improve performance by providing richer historical context, the gains diminish beyond certain thresholds, suggesting a trade-off between context richness and effective information processing. This pattern is consistent across all evaluation categories, indicating the importance of balanced context retrieval for optimal performance.

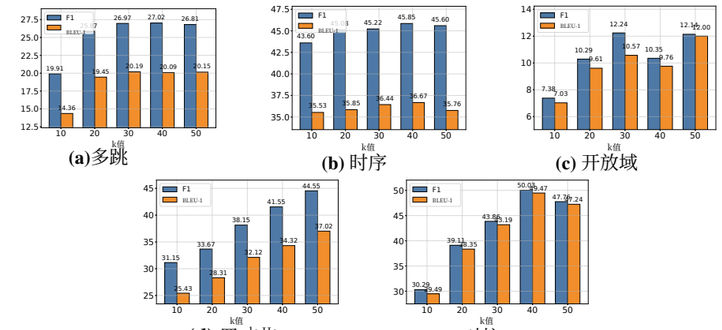
Table 4: Comparison of memory usage and retrieval time across different memory methods and scales.

Memory Size	Method	Memory Usage (MB)	Retrieval Time (μs)
1,000	A-MEM	1.46	0.31 ± 0.30
	MemoryBank [39]	1.46	0.24 ± 0.20
	ReadAgent [17]	1.46	43.62 ± 8.47
10,000	A-MEM	14.65	0.38 ± 0.25
	MemoryBank [39]	14.65	0.26 ± 0.13
	ReadAgent [17]	14.65	484.45 ± 93.86
100,000	A-MEM	146.48	1.40 ± 0.49
	MemoryBank [39]	146.48	0.78 ± 0.26
	ReadAgent [17]	146.48	6,682.22 ± 111.63
1,000,000	A-MEM	1464.84	3.70 ± 0.74
	MemoryBank [39]	1464.84	1.91 ± 0.31
	ReadAgent [17]	1464.84	120,069.68 ± 1,673.39

tasks. The observation suggests a delicate balance in memory retrieval - while larger k values provide richer historical context for reasoning, they may also introduce noise and challenge the model's capacity to process longer sequences effectively. Our analysis indicates that moderate k values strike an optimal balance between context richness and information processing efficiency.

4.6 Scaling Analysis

To evaluate storage costs with accumulating memory, we examined the relationship between storage size and retrieval time across our A-MEM system and two baseline approaches: MemoryBank [39] and ReadAgent [17]. We evaluated these three memory systems with identical memory content across four scale points, increasing the number of entries by a factor of 10 at each step (from 1,000 to 10,000, 100,000, and finally 1,000,000 entries). The experimental results reveal key insights about our A-MEM system's scaling properties: In terms of space complexity, all three systems exhibit identical linear memory usage scaling ($O(N)$), as expected for vector-based retrieval systems. This confirms that A-MEM introduces no additional storage overhead compared to baseline approaches. For retrieval time, A-MEM demonstrates excellent efficiency with minimal increases as memory size grows. Even when scaling to 1 million memories, A-MEM's retrieval time increases only from 0.31μs to 3.70μs, representing exceptional performance. While MemoryBank shows slightly faster retrieval times, A-MEM maintains comparable performance while providing richer memory representations and functionality. Based on our space complexity and retrieval time analysis, we conclude that A-MEM's retrieval mechanisms maintain excellent efficiency even at large scales. The minimal growth in retrieval time across memory sizes addresses concerns about efficiency in large-scale memory systems, demonstrating that A-MEM provides a highly scalable solution for long-term conversation management. This unique combination of efficiency, scalability, and enhanced memory capabilities positions A-MEM as a significant advancement in building powerful and long-term memory mechanism for LLM Agents.



(a)多跳 **(b)** 时序 **(c)** 开放域

(d) 罪$g^{le\ Ho}$ p **(e)** 对抗

图3： 不同任务类别中记忆检索参数k对性能的影响（以GPT-4o-mini为基础模型）。虽然较大的k值通过提供更丰富的历史上下文通常会提升性能，但收益在超过特定阈值后会逐渐减少，表明上下文丰富度与有效信息处理之间存在权衡。这种模式在所有评估类别中均保持一致，表明平衡的上下文检索对最佳性能至关重要。

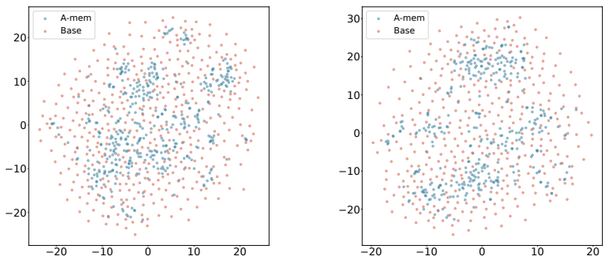
表4： 不同记忆方法及规模下的内存使用和检索时间对比 s.

内存大小	方法	内存使用量（MB）	检索时间（μs）
1000	A-MEM	1.46	0.31 ± 0.30
	MEMORYBANK [39]	1.46	0.24 ± 0.20
	READAGENT [17]	1.46	43.62 ± 8.47
10000	A-MEM	14.65	0.38 ± 0.25
	MEMORYBANK [39]	14.65	0.26 ± 0.13
	READAGENT [17]	14.65	484.45 ± 93.86
100,000	A-MEM	146.48	1.40 ± 0.49
	MEMORYBANK [39]	146.48	0.78 ± 0.26
	READAGENT [17]	146.48	6,682.22 ± 111.63
1,000,000	A-MEM	1464.84	3.70 ± 0.74
	MEMORYBANK [39]	1464.84	1.91 ± 0.31
	读取 [17]	1464.84	120,069.68 ± 1,673.39

任务中。观察结果表明，记忆检索存在一种微妙的平衡——较大的k值提供了更丰富的推理历史上下文，但也可能引入噪声并挑战模型处理长序列的能力。我们的分析表明，适中的k值在上下文丰富度和信息处理效率之间取得了最佳平衡。

4.6 扩展分析

为评估累积记忆的存储成本，我们考察了存储大小与检索时间之间的关系，对象是我们A-MEM 系统以及两种基线方法：MEMORYBANK [39]和READAGENT [17]。我们在四个规模点（规模点）上评估了这三个记忆系统，内存内容相同，每一步将条目数量增加10倍（从1,000到10,000、100,000，最终到1,000,000条条目）。实验结果揭示了我们A-MEM系统扩展特性的关键洞察：就空间复杂度而言，所有三个系统均表现出预期的、相同的线性内存使用缩放($O(N)$)，这是基于向量的检索系统的特点。这证实了与基线方法相比，A-MEM不会引入额外的存储开销。在检索时间方面，A-MEM展现出卓越的效率，随着内存大小的增长，检索时间仅略有增加。即使扩展到1百万条记忆时，A-MEM的检索时间也仅从0.31μs增加到3.70μs，代表了优异的性能。虽然MEMORYBANK的检索时间略快，但A-MEM在保持相当性能的同时，提供了更丰富的记忆表示和功能。基于我们的空间复杂度和检索时间分析，我们得出结论：A-MEM的检索机制即使在大规模下也能保持极高的效率。检索时间在内存大小上的最小增长解决了大规模记忆系统效率的担忧，证明A-MEM为长期对话管理提供了一种高度可扩展的解决方案。这种效率、可扩展性和增强记忆能力的独特结合，使A-MEM成为构建强大和长期记忆机制的重要进步，适用于大语言模型代理。



(a) Dialogue 1 **(b)** Dialogue 2

Figure 4: T-SNE Visualization of Memory Embeddings Showing More Organized Distribution with A-MEM (blue) Compared to Base Memory (red) Across Different Dialogues. Base Memory represents A-MEM without link generation and memory evolution.

4.7 Memory Analysis

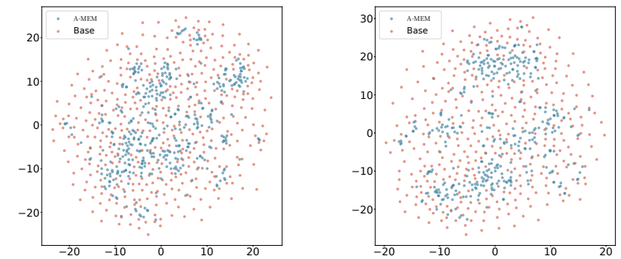
We present the t-SNE visualization in Figure 4 of memory embeddings to demonstrate the structural advantages of our agentic memory system. Analyzing two dialogues sampled from long-term conversations in LoCoMo [22], we observe that A-MEM (shown in blue) consistently exhibits more coherent clustering patterns compared to the baseline system (shown in red). This structural organization is particularly evident in Dialogue 2, where well-defined clusters emerge in the central region, providing empirical evidence for the effectiveness of our memory evolution mechanism and contextual description generation. In contrast, the baseline memory embeddings display a more dispersed distribution, demonstrating that memories lack structural organization without our link generation and memory evolution components. These visualization results validate that A-MEM can autonomously maintain meaningful memory structures through dynamic evolution and linking mechanisms. More results can be seen in Appendix A.4.

5 Conclusions

In this work, we introduced A-MEM, a novel agentic memory system that enables LLM agents to dynamically organize and evolve their memories without relying on predefined structures. Drawing inspiration from the Zettelkasten method, our system creates an interconnected knowledge network through dynamic indexing and linking mechanisms that adapt to diverse real-world tasks. The system's core architecture features autonomous generation of contextual descriptions for new memories and intelligent establishment of connections with existing memories based on shared attributes. Furthermore, our approach enables continuous evolution of historical memories by incorporating new experiences and developing higher-order attributes through ongoing interactions. Through extensive empirical evaluation across six foundation models, we demonstrated that A-MEM achieves superior performance compared to existing state-of-the-art baselines in long-term conversational tasks. Visualization analysis further validates the effectiveness of our memory organization approach. These results suggest that agentic memory systems can significantly enhance LLM agents' ability to utilize long-term knowledge in complex environments.

6 Limitations

While our agentic memory system achieves promising results, we acknowledge several areas for potential future exploration. First, although our system dynamically organizes memories, the quality of these organizations may still be influenced by the inherent capabilities of the underlying language models. Different LLMs might generate slightly different contextual descriptions or establish varying connections between memories. Additionally, while our current implementation focuses on text-based interactions, future work could explore extending the system to handle multimodal information, such as images or audio, which could provide richer contextual representations.



(a)对话 1 **(b)** 对话 2

图 4： 记忆嵌入的 t-SNE 可视化，显示 A-MEM（蓝色）与基础记忆（红色）在不同对话中分布更组织化。基础记忆代表没有链接生成和记忆进化的 A-MEM。

4.7 记忆分析

我们在图4中展示了t-SNE可视化，以演示我们能动记忆系统的结构优势。分析从LoCoMo长期对话中采样的两个对话，我们观察到A-MEM (以蓝色显示)与基线系统（以红色显示）相比，始终表现出更连贯的聚类模式。这种结构组织在对话2中尤为明显，在中心区域形成了界限清晰的聚类，为我们的记忆进化机制和上下文描述生成提供了经验证据。相比之下，基线记忆嵌入呈现出更分散的分布，表明在没有我们的链接生成和记忆进化组件的情况下，记忆缺乏结构组织。这些可视化结果证实，A-MEM可以通过动态进化和链接机制自主地维持有意义的记忆结构。更多结果可以在附录A.4中看到。

5 结论

在这项工作中，我们介绍了 A-MEM，一种新颖的能动记忆系统，它使大语言模型代理能够在不依赖预定义结构的情况下动态组织和进化其记忆。受卡片盒方法的启发，我们的系统通过适应不同现实任务的动态索引和链接机制创建了一个相互连接的知识网络。该系统的核心架构特征在于能够自主生成新记忆的上下文描述，并能基于共享属性智能地与现有记忆建立连接。此外，我们的方法通过结合新经历和通过持续的交互发展高阶属性，使历史记忆得以持续进化。通过在六个基础模型上的广泛实证评估，我们证明了 A-MEM 在长期对话任务中比现有最先进基线实现了更优越的性能。可视化分析进一步验证了我们记忆组织方法的有效性。这些结果表明，能动记忆系统可以显著增强大语言模型代理在复杂环境中利用长期知识的能力。

6 局限性

尽管我们的能动记忆系统能够取得有前景的结果，但我们承认存在一些潜在的未来探索领域。首先，尽管我们的系统能够动态组织记忆，但这些组织的质量可能仍会受到底层语言模型固有能力的制约。不同的LLM可能会生成略有不同的上下文描述，或在记忆之间建立不同的连接。此外，虽然我们当前的实现专注于基于文本的交互，未来的工作可以探索将系统扩展到处理多模态信息，例如图像或音频，这可以提供更丰富的上下文表示。

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APPENDIX

A Experiment

A.1 Detailed Baselines Introduction

LoCoMo [22] takes a direct approach by leveraging foundation models without memory mechanisms for question answering tasks. For each query, it incorporates the complete preceding conversation and questions into the prompt, evaluating the model's reasoning capabilities.

ReadAgent [17] tackles long-context document processing through a sophisticated three-step methodology: it begins with episode pagination to segment content into manageable chunks, followed by memory gisting to distill each page into concise memory representations, and concludes with interactive look-up to retrieve pertinent information as needed.

MemoryBank [39] introduces an innovative memory management system that maintains and efficiently retrieves historical interactions. The system features a dynamic memory updating mechanism based on the Ebbinghaus Forgetting Curve theory, which intelligently adjusts memory strength according to time and significance. Additionally, it incorporates a user portrait building system that progressively refines its understanding of user personality through continuous interaction analysis.

MemGPT [25] presents a novel virtual context management system drawing inspiration from traditional operating systems' memory hierarchies. The architecture implements a dual-tier structure: a main context (analogous to RAM) that provides immediate access during LLM inference, and an external context (analogous to disk storage) that maintains information beyond the fixed context window.

A.2 Evaluation Metric

The F1 score represents the harmonic mean of precision and recall, offering a balanced metric that combines both measures into a single value. This metric is particularly valuable when we need to balance between complete and accurate responses:

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{11}$$

where

$$\text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \tag{12}$$

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \tag{13}$$

In question-answering systems, the F1 score serves a crucial role in evaluating exact matches between predicted and reference answers. This is especially important for span-based QA tasks, where systems must identify precise text segments while maintaining comprehensive coverage of the answer.

BLEU-1 [26] provides a method for evaluating the precision of unigram matches between system outputs and reference texts:

$$\text{BLEU-1} = BP \cdot \exp\left(\sum_{n=1}^{1} w_n \log p_n\right) \tag{14}$$

where

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases} \tag{15}$$

$$p_n = \frac{\sum_i \sum_k \min(h_{ik}, m_{ik})}{\sum_i \sum_k h_{ik}} \tag{16}$$

附录

A 实验

A.1 详细基线介绍

LOCOMO [22] 采用直接方法，通过利用没有记忆机制的基础模型来处理问答任务。对于每个查询，它将完整的先前对话和问题整合到提示中，评估模型的推理能力。

读取代理 [17] 通过复杂的三个步骤方法处理长上下文文档：它首先使用场景分页将内容分割成可管理的块，然后通过记忆梗概将每一页提炼成简洁的记忆表示，最后通过交互式查找按需检索相关信息。

记忆库 [39] 引入了一种创新的记忆管理系统，用于维护和高效检索历史交互。该系统基于艾宾浩斯遗忘曲线理论，采用动态记忆更新机制，根据时间和重要性智能调整记忆强度。此外，它还包含一个用户画像构建系统，通过持续交互分析逐步完善对用户个性的理解。

MemGPT [25] 提出了一种借鉴传统操作系统内存层次结构的虚拟上下文管理系统。该架构采用双层结构：主上下文（类似于RAM）在LLM推理时提供即时访问，以及外部上下文（类似于磁盘存储）维护固定上下文窗口之外的信息。

A.2 评估指标

F1分数表示精确率和召回率的调和平均值，是一种平衡的指标，将这两种度量结合为一个值。当我们需要在完整和准确的响应之间取得平衡时，这种指标特别有价值：

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{11}$$

其中

$$\text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \tag{12}$$

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \tag{13}$$

在问答系统中，F1分数在评估预测答案与参考答案之间的精确匹配方面起着关键作用。这对于基于片段的问答任务尤其重要，系统必须识别精确的文本片段，同时保持对答案的全面覆盖。

BLEU-1 [26] 提供了一种评估系统输出与参考文本之间单词匹配精度的方法：

$$\text{BLEU-1} = BP \cdot \exp\left(\sum_{n=1}^{1} w_n \log p_n\right) \tag{14}$$

哪里

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases} \tag{15}$$

$$p_n = \frac{\sum_i \sum_k \min(h_{ik}, m_{ik})}{\sum_i \sum_k h_{ik}} \tag{16}$$

Here, c is candidate length, r is reference length, h_{ik} is the count of n-gram i in candidate k, and m_{ik} is the maximum count in any reference. In QA, BLEU-1 evaluates the lexical precision of generated answers, particularly useful for generative QA systems where exact matching might be too strict.

ROUGE-L [19] measures the longest common subsequence between the generated and reference texts.

$$\text{ROUGE-L} = \frac{(1+\beta^2)R_l P_l}{R_l + \beta^2 P_l} \tag{17}$$

$$R_l = \frac{\text{LCS}(X,Y)}{|X|} \tag{18}$$

$$P_l = \frac{\text{LCS}(X,Y)}{|Y|} \tag{19}$$

where X is reference text, Y is candidate text, and LCS is the Longest Common Subsequence.

ROUGE-2 [19] calculates the overlap of bigrams between the generated and reference texts.

$$\text{ROUGE-2} = \frac{\sum_{\text{bigram}\in\text{ref}} \min(\text{Count}_{\text{ref}}(\text{bigram}), \text{Count}_{\text{cand}}(\text{bigram}))}{\sum_{\text{bigram}\in\text{ref}} \text{Count}_{\text{ref}}(\text{bigram})} \tag{20}$$

Both ROUGE-L and ROUGE-2 are particularly useful for evaluating the fluency and coherence of generated answers, with ROUGE-L focusing on sequence matching and ROUGE-2 on local word order.

METEOR [5] computes a score based on aligned unigrams between the candidate and reference texts, considering synonyms and paraphrases.

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}) \tag{21}$$

$$F_{\text{mean}} = \frac{10P \cdot R}{R + 9P} \tag{22}$$

$$\text{Penalty} = 0.5 \cdot (\frac{\text{ch}}{m})^3 \tag{23}$$

where P is precision, R is recall, ch is number of chunks, and m is number of matched unigrams. METEOR is valuable for QA evaluation as it considers semantic similarity beyond exact matching, making it suitable for evaluating paraphrased answers.

SBERT Similarity [27] measures the semantic similarity between two texts using sentence embeddings.

$$\text{SBERT_Similarity} = \cos(\text{SBERT}(x), \text{SBERT}(y)) \tag{24}$$

$$\cos(a,b) = \frac{a \cdot b}{||a||||b||} \tag{25}$$

SBERT(x) represents the sentence embedding of text. SBERT Similarity is particularly useful for evaluating semantic understanding in QA systems, as it can capture meaning similarities even when the lexical overlap is low.

A.3 Comparison Results

Our comprehensive evaluation using ROUGE-2, ROUGE-L, METEOR, and SBERT metrics demonstrates that A-MEM achieves superior performance while maintaining remarkable computational efficiency. Through extensive empirical testing across various model sizes and task categories, we have established A-MEM as a more effective approach compared to existing baselines, supported by several compelling findings. In our analysis of non-GPT models, specifically Qwen2.5 and Llama 3.2, A-MEM consistently outperforms all baseline approaches across all metrics. The Multi-Hop category showcases particularly striking results, where Qwen2.5-15b with A-MEM achieves a ROUGE-L score of 27.23, dramatically surpassing LoComo's 4.68 and ReadAgent's 2.81 - representing a nearly six-fold improvement. This pattern of superiority extends consistently across METEOR and SBERT

这里， c 是候选长度， r 是参考长度， h_{ik} 是候选中n-gram i的出现次数，而 m_{ik}是任何参考中的最大计数。在问答中，BLEU-1评估生成答案的词汇精确度，特别适用于生成式问答系统，因为精确匹配可能过于严格。

ROUGE-L [19] 测量生成文本和参考文本之间的最长公共子序列
.文本

$$\text{ROUGE-L} = \frac{(1+\beta^2)R_l P_l}{R_l + \beta^2 P_l} \tag{17}$$

$$R_l = \frac{\text{LCS}(X,Y)}{|X|} \tag{18}$$

$$P_l = \frac{\text{LCS}(X,Y)}{|Y|} \tag{19}$$

其中 X 是参考文本， Y 是候选文本，LCS 是最长公共子序列 nce.

ROUGE-2 [19] 计算生成文本和参考文本之间的大二元组的重叠度 .

$$\text{ROUGE-2} = \frac{\sum_{\text{bigram}\in\text{ref}} \min(\text{Count}_{\text{ref}}(\text{bigram}), \text{Count}_{\text{cand}}(\text{bigram}))}{\sum_{\text{bigram}\in\text{ref}} \text{Count}_{\text{ref}}(\text{bigram})} \tag{20}$$

ROUGE-L 和 ROUGE-2 都特别适用于评估生成答案的流畅性和连贯性，其中 ROUGE-L 关注序列匹配，ROUGE-2 关注局部词序

METEOR [5] 基于候选文本和参考文本之间对齐的单元词计算得分，考虑同义词和释义

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}) \tag{21}$$

$$F_{\text{mean}} = \frac{10P \cdot R}{R + 9P} \tag{22}$$

$$\text{Penalty} = 0.5 \cdot (\frac{\text{ch}}{m})^3 \tag{23}$$

P 是精确率， R 是召回率，ch 是块数， m 是匹配到的单字数。METEOR 对于问答评估很有价值，因为它考虑了超越精确匹配的语义相似性，因此适合评估释义答案。

SBERT Similarity [27] 使用句子嵌入来衡量两个文本之间的语义相似度。

$$\text{SBERT_Similarity} = \cos(\text{SBERT}(x), \text{SBERT}(y)) \tag{24}$$

$$\cos(a,b) = \frac{a \cdot b}{||a||||b||} \tag{25}$$

SBERT(x) 代表文本的句子嵌入。SBERT Similarity 对于评估问答系统中的语义理解特别有用，因为它即使当词汇重叠较低时也能捕捉到意义相似性。

A.3 比较结果

我们使用ROUGE-2、ROUGE-L、METEOR和SBERT指标进行的全面评估表明，A-MEM在保持卓越计算效率的同时实现了优异性能。通过在各种模型大小和任务类别中进行广泛的实证测试，我们已将A-MEM确立为比现有基线更有效的方法，并得到了若干有力发现的支撑。在我们的非GPT模型分析中，特别是针对Qwen2.5和Llama 3.2，A-MEM在所有指标上始终优于所有基线方法。多跳类别展示了尤为突出的结果，其中Qwen2.5-15b配合A-MEM实现了27.23的ROUGE-L得分，大幅超越了LoComo的4.68和ReadAgent的2.81——这代表近六倍的提升。这种优势在METEOR和SBERT指标上持续显现。

Table 5: Experimental results on LoCoMo dataset of QA tasks across five categories (Multi Hop, Temporal, Open Domain, Single Hop, and Adversial) using different methods. Results are reported in ROUGE-2 and ROUGE-L scores, abbreviated to RGE-2 and RGE-L. The best performance is marked in bold, and our proposed method A-MEM (highlighted in gray) demonstrates competitive performance across six foundation language models.

Model		Method	Category									
			Multi Hop		Temporal		Open Domain		Single Hop		Adversial	
			RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L
GPT	4o-mini	LoCoMo	9.64	23.92	2.01	18.09	3.40	11.58	26.48	40.20	**60.46**	**69.59**
		ReadAgent	2.47	9.45	0.95	13.12	0.55	5.76	2.99	9.92	6.66	9.79
		MemoryBank	1.18	5.43	0.52	9.64	0.97	5.77	1.64	6.63	4.55	7.35
		MemGPT	10.58	25.60	4.76	25.22	0.76	9.14	28.44	42.24	36.62	43.75
		A-Mem	**10.61**	**25.86**	**21.39**	**44.27**	**3.42**	**12.09**	**29.50**	**45.18**	42.62	50.04
	4o	LoCoMo	11.53	30.65	1.68	8.17	3.21	16.33	**45.42**	**63.86**	**45.13**	**52.67**
		ReadAgent	3.91	14.36	0.43	3.96	0.52	8.58	4.75	13.41	4.24	6.81
		MemoryBank	1.84	7.36	0.36	2.29	2.13	6.85	3.02	9.35	1.22	4.41
		MemGPT	11.55	30.18	4.66	15.83	3.27	14.02	43.27	62.75	28.72	35.08
		A-Mem	**12.76**	**31.71**	**9.82**	**25.04**	**6.09**	**16.63**	33.67	50.31	30.31	36.34
Qwen2.5	1.5b	LoCoMo	1.39	9.24	0.00	4.68	3.42	10.59	3.25	11.15	35.10	43.61
		ReadAgent	0.74	7.14	0.10	2.81	3.05	12.63	1.47	7.88	20.73	27.82
		MemoryBank	1.51	11.18	0.14	5.39	1.80	8.44	5.07	13.72	29.24	36.95
		MemGPT	1.16	11.35	0.00	7.88	2.87	14.62	2.18	9.82	23.96	31.69
		A-Mem	**4.88**	**17.94**	**5.88**	**27.23**	**3.44**	**16.87**	**12.32**	**24.38**	**36.32**	**46.60**
	3b	LoCoMo	0.49	4.83	0.14	3.20	1.31	5.38	1.97	6.98	12.66	17.10
		ReadAgent	0.08	4.08	0.00	1.96	1.26	6.19	0.73	4.34	7.35	10.64
		MemoryBank	0.43	3.76	0.05	1.61	0.24	6.32	1.03	4.22	9.55	13.41
		MemGPT	0.69	5.55	0.05	3.17	1.90	7.90	2.05	7.32	10.46	14.39
		A-Mem	**2.91**	**12.42**	**8.11**	**27.74**	**1.51**	**7.51**	**8.80**	**17.57**	**21.39**	**27.98**
Llama 3.2	1b	LoCoMo	2.51	11.48	0.44	8.25	1.69	13.06	2.94	13.00	39.85	52.74
		ReadAgent	0.53	6.49	0.00	4.62	5.47	14.29	1.19	8.03	34.52	45.55
		MemoryBank	2.96	13.57	0.23	10.53	4.01	18.38	6.41	17.66	41.15	53.31
		MemGPT	1.82	9.91	0.06	6.56	2.13	11.36	2.00	10.37	38.59	50.31
		A-Mem	**4.82**	**19.31**	**1.84**	**20.47**	**5.99**	**18.49**	**14.82**	**29.78**	**46.76**	**60.23**
	3b	LoCoMo	0.98	7.22	0.03	4.45	2.36	11.39	2.85	8.45	25.47	30.26
		ReadAgent	2.47	1.78	3.01	3.01	5.07	5.22	3.25	2.51	15.78	14.01
		MemoryBank	1.83	6.96	0.25	3.41	0.43	4.43	2.73	7.83	14.64	18.59
		MemGPT	0.72	5.39	0.11	2.85	0.61	5.74	1.45	4.42	16.62	21.47
		A-Mem	**6.02**	**17.62**	**7.93**	**27.97**	**5.38**	**13.00**	**16.89**	**28.55**	**35.48**	**42.25**

scores. When examining GPT-based models, our results reveal an interesting pattern. While LoCoMo and MemGPT demonstrate strong capabilities in Open Domain and Adversarial tasks, A-Mem shows remarkable superiority in Multi-Hop reasoning tasks. Using GPT-4o-mini, A-Mem achieves a ROUGE-L score of 44.27 in Multi-Hop tasks, more than doubling LoCoMo's 18.09. This significant advantage maintains consistency across other metrics, with METEOR scores of 23.43 versus 7.61 and SBERT scores of 70.49 versus 52.30. The significance of these results is amplified by A-Mem's exceptional computational efficiency. Our approach requires only 1,200-2,500 tokens, compared to the substantial 16,900 tokens needed by LoCoMo and MemGPT. This efficiency stems from two key architectural innovations: First, our novel agentic memory architecture creates interconnected memory networks through atomic notes with rich contextual descriptions, enabling more effective capture and utilization of information relationships. Second, our selective top-k retrieval mechanism facilitates dynamic memory evolution and structured organization. The effectiveness of these innovations is particularly evident in complex reasoning tasks, as demonstrated by the consistently strong Multi-Hop performance across all evaluation metrics. Besides, we also show the experimental results with different foundational models including DeepSeek-R1-32B [11], Claude 3.0 Haiku [2] and Claude 3.5 Haiku [3].

A.4 Memory Analysis

In addition to the memory visualizations of the first two dialogues shown in the main text, we present additional visualizations in Fig.5 that demonstrate the structural advantages of our agentic memory system. Through analysis of two dialogues sampled from long-term conversations in LoCoMo[22], we observe that A-Mem (shown in blue) consistently produces more coherent clustering patterns compared to the baseline system (shown in red). This structural organization is particularly evident in Dialogue 2, where distinct clusters emerge in the central region, providing empirical support for the effectiveness of our memory evolution mechanism and contextual description generation. In contrast, the baseline memory embeddings exhibit a more scattered distribution, indicating that memories lack structural organization without our link generation and memory evolution components.

表5： 在LoCoMo数据集上针对问答任务的五个类别（多跳、时序、开放域、单跳和对抗）使用不同方法进行的实验结果。结果以ROUGE-2和ROUGE-L分数报告，缩写为RGE-2和RGE-L。最佳性能以粗体标记，我们提出的方法A-MEM（以灰色突出显示）在六个基础语言模型上表现出具有竞争力的性能。

模型		方法	分类									
			多跳		时序		开放域		单跳		对抗	
			RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L	RGE-2	RGE-L
GPT	4o-mini	LOCOMO	9.64	23.92	2.01	18.09	3.40	11.58	26.48	40.20	**60.46**	**69.59**
		READAGENT	2.47	9.45	0.95	13.12	0.55	5.76	2.99	9.92	6.66	9.79
		MEMORYBANK	1.18	5.43	0.52	9.64	0.97	5.77	1.64	6.63	4.55	7.35
		MEMGPT	10.58	25.60	4.76	25.22	0.76	9.14	28.44	42.24	36.62	43.75
		A-MEM	**10.61**	**25.86**	**21.39**	**44.27**	**3.42**	**12.09**	**29.50**	**45.18**	42.62	50.04
	4o	LOCOMO	11.53	30.65	1.68	8.17	3.21	16.33	**45.42**	**63.86**	**45.13**	**52.67**
		READAGENT	3.91	14.36	0.43	3.96	0.52	8.58	4.75	13.41	4.24	6.81
		MEMORYBANK	1.84	7.36	0.36	2.29	2.13	6.85	3.02	9.35	1.22	4.41
		MEMGPT	11.55	30.18	4.66	15.83	3.27	14.02	43.27	62.75	28.72	35.08
		A-MEM	**12.76**	**31.71**	**9.82**	**25.04**	**6.09**	**16.63**	33.67	50.31	30.31	36.34
Qwen2.5	1.5b	LOCOMO	1.39	9.24	0.00	4.68	3.42	10.59	3.25	11.15	35.10	43.61
		READAGENT	0.74	7.14	0.10	2.81	3.05	12.63	1.47	7.88	20.73	27.82
		MEMORYBANK	1.51	11.18	0.14	5.39	1.80	8.44	5.07	13.72	29.24	36.95
		MEMGPT	1.16	11.35	0.00	7.88	2.87	14.62	2.18	9.82	23.96	31.69
		A-MEM	**4.88**	**17.94**	**5.88**	**27.23**	**3.44**	**16.87**	**12.32**	**24.38**	**36.32**	**46.60**
	3b	LOCOMO	0.49	4.83	0.14	3.20	1.31	5.38	1.97	6.98	12.66	17.10
		READAGENT	0.08	4.08	0.00	1.96	1.26	6.19	0.73	4.34	7.35	10.64
		MEMORYBANK	0.43	3.76	0.05	1.61	0.24	6.32	1.03	4.22	9.55	13.41
		MEMGPT	0.69	5.55	0.05	3.17	1.90	7.90	2.05	7.32	10.46	14.39
		A-MEM	**2.91**	**12.42**	**8.11**	**27.74**	**1.51**	**7.51**	**8.80**	**17.57**	**21.39**	**27.98**
Llama 3.2	1b	LOCOMO	2.51	11.48	0.44	8.25	1.69	13.06	2.94	13.00	39.85	52.74
		READAGENT	0.53	6.49	0.00	4.62	5.47	14.29	1.19	8.03	34.52	45.55
		MEMORYBANK	2.96	13.57	0.23	10.53	4.01	18.38	6.41	17.66	41.15	53.31
		MEMGPT	1.82	9.91	0.06	6.56	2.13	11.36	2.00	10.37	38.59	50.31
		A-MEM	**4.82**	**19.31**	**1.84**	**20.47**	**5.99**	**18.49**	**14.82**	**29.78**	**46.76**	**60.23**
	3b	LOCOMO	0.98	7.22	0.03	4.45	2.36	11.39	2.85	8.45	25.47	30.26
		READAGENT	2.47	1.78	3.01	3.01	5.07	5.22	3.25	2.51	15.78	14.01
		MEMORYBANK	1.83	6.96	0.25	3.41	0.43	4.43	2.73	7.83	14.64	18.59
		MEMGPT	0.72	5.39	0.11	2.85	0.61	5.74	1.45	4.42	16.62	21.47
		A-MEM	**6.02**	**17.62**	**7.93**	**27.97**	**5.38**	**13.00**	**16.89**	**28.55**	**35.48**	**42.25**

得分。当考察基于GPT的模型时，我们的结果揭示了一种有趣的模式。虽然LoComo和MemGPT在开放域和对抗性任务中表现出强大能力，但A-Mem在多跳推理任务中展现出卓越优势。使用GPT-4o-mini，A-Mem 在多跳任务中实现了44.27的ROUGE-L得分，是LoComo的18.09的两倍多。这种显著优势在METEOR得分（23.43 vs 7.61）和SBERT得分（70.49 vs 52.30）等其他指标上同样保持一致。这些结果的显著性因A-Mem的非凡计算效率而进一步放大。我们的方法仅需1,200-2,500个token，而LoComo和MemGPT则需要大量的16,900个token。这种效率源于两项关键架构创新：首先，我们创新的能动式记忆架构通过原子笔记和丰富的上下文描述创建互联的记忆网络，从而更有效地捕捉和利用信息关系。其次，我们选择性的top-k检索机制促进了动态记忆进化和结构化组织。这些创新的效能在复杂推理任务中尤为明显，如所有评估指标上持续优异的多跳性能所示。此外，我们还展示了不同基础模型（包括DeepSeek-R1-32B [11], Claude 3.0 Haiku [2]和Claude 3.5 Haiku [3]）的实验结果。

A.4 记忆分析

除了在正文中展示的前两个对话的记忆可视化之外，我们在图<样式 id='1'>5</样式><样式 id='3'>中展示了其他可视化，以证明我们的能动记忆系统的结构优势。通过分析从LOCOMO长期对话中采样的两个对话<样式 id='3'>[22],，我们观察到A-MEM（蓝色显示）与基线系统（红色显示）相比，始终能产生更连贯的聚类模式。这种结构组织在对话2中尤为明显，其中在中心区域出现了不同的聚类，为我们的记忆进化机制和上下文描述生成提供了实证支持。相比之下，基线记忆嵌入表现出更分散的分布，表明在没有我们的链接生成和记忆进化组件的情况下，记忆缺乏结构组织。

Table 6: Experimental results on LoCoMo dataset of QA tasks across five categories (Multi Hop, Temporal, Open Domain, Single Hop, and Adversial) using different methods. Results are reported in METEOR and SBERT Similarity scores, abbreviated to ME and SBERT. The best performance is marked in bold, and our proposed method A-MEM (highlighted in gray) demonstrates competitive performance across six foundation language models.

Model		Method	Category									
			Multi Hop		Temporal		Open Domain		Single Hop		Adversial	
			ME	SBERT	ME	SBERT	ME	SBERT	ME	SBERT	ME	SBERT
GPT	4o-mini	LoCoMo	15.81	47.97	7.61	52.30	8.16	35.00	40.42	57.78	**63.28**	**71.93**
		ReadAgent	5.46	28.67	4.76	45.07	3.69	26.72	8.01	26.78	8.38	15.20
		MemoryBank	3.42	21.71	4.07	37.58	4.21	23.71	5.81	20.76	6.24	13.00
		MemGPT	15.79	49.33	13.25	61.53	4.59	32.77	41.40	58.19	39.16	47.24
		A-Mem	**16.36**	**49.46**	**23.43**	**70.49**	**8.36**	**38.48**	**42.32**	**59.38**	45.64	53.26
	4o	LoCoMo	16.34	53.82	7.21	32.15	8.98	**43.72**	**53.39**	**73.40**	**47.72**	**56.09**
		ReadAgent	7.86	37.41	3.76	26.22	4.42	30.75	9.36	31.37	5.47	12.34
		MemoryBank	3.22	26.23	2.29	23.49	4.18	24.89	6.64	23.90	2.93	10.01
		MemGPT	16.64	55.12	12.68	35.93	7.78	37.91	52.14	72.83	31.15	39.08
		A-Mem	**17.53**	**55.96**	**13.10**	**45.40**	**10.62**	38.87	41.93	62.47	32.34	40.11
Qwen2.5	1.5b	LoCoMo	4.99	32.23	2.86	34.03	5.89	35.61	8.57	29.47	40.53	50.49
		ReadAgent	3.67	28.20	1.88	27.27	8.97	35.13	5.52	26.33	24.04	34.12
		MemoryBank	5.57	35.40	2.80	32.47	4.27	33.85	10.59	32.16	32.93	42.83
		MemGPT	5.40	35.64	2.35	39.04	7.68	40.36	7.07	30.16	27.24	40.63
		A-Mem	**9.49**	**43.49**	**11.92**	**61.65**	**9.11**	**42.58**	**19.69**	**41.93**	**40.64**	**52.44**
	3b	LoCoMo	2.00	24.37	1.92	25.24	3.45	25.38	6.00	21.28	16.67	23.14
		ReadAgent	1.78	21.10	1.69	20.78	4.43	25.15	3.37	18.20	10.46	17.39
		MemoryBank	2.37	17.81	2.22	21.93	3.86	20.65	3.99	16.26	15.49	20.77
		MemGPT	3.74	24.31	2.25	27.67	6.44	29.59	6.24	22.40	13.19	20.83
		A-Mem	**6.25**	**33.72**	**14.04**	**62.54**	**6.56**	**30.60**	**15.98**	**33.98**	**27.36**	**33.72**
Llama 3.2	1b	LoCoMo	5.77	38.02	3.38	45.44	6.20	42.69	9.33	34.19	46.79	60.74
		ReadAgent	2.97	29.26	1.31	26.45	7.13	39.19	5.36	26.44	42.39	54.35
		MemoryBank	6.77	39.33	4.43	45.63	7.76	42.81	13.01	37.32	50.43	60.81
		MemGPT	5.10	32.99	2.54	41.81	3.26	35.99	6.62	30.68	45.00	61.33
		A-Mem	**9.01**	**45.16**	**7.50**	**54.79**	**8.30**	**43.42**	**22.46**	**47.07**	**53.72**	**68.00**
	3b	LoCoMo	3.69	27.94	2.96	20.40	6.46	32.17	6.58	22.92	29.02	35.74
		ReadAgent	1.21	17.40	2.33	12.02	3.39	19.63	2.46	14.63	14.37	21.25
		MemoryBank	3.84	25.06	2.73	13.65	3.05	21.08	6.35	22.02	17.14	24.39
		MemGPT	2.78	22.06	2.21	14.97	3.63	23.18	3.47	17.81	20.50	26.87
		A-Mem	**9.74**	**39.32**	**13.19**	**59.70**	**8.09**	**32.27**	**24.30**	**42.86**	**39.74**	**46.76**

Table 7: Experimental results on LoCoMo dataset of QA tasks across five categories (Multi Hop, Temporal, Open Domain, Single Hop, and Adversial) using different methods. Results are reported in F1 and BLEU-1 (%) scores with different foundation models.

Method	Category									
	Multi Hop		Temporal		Open Domain		Single Hop		Adversial	
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
DeepSeek-R1-32B										
LoCoMo	8.58	6.48	4.79	4.35	12.96	12.52	10.72	8.20	21.40	20.23
MemGPT	8.28	6.25	5.45	4.97	10.97	9.09	11.34	9.03	**30.77**	**29.23**
A-Mem	**15.02**	**10.64**	**14.64**	**11.01**	**14.81**	**12.82**	**15.37**	**12.30**	27.92	27.19
Claude 3.0 Haiku										
LoCoMo	4.56	3.33	0.82	0.59	2.86	3.22	3.56	3.24	3.46	3.42
MemGPT	7.65	6.36	1.65	1.26	7.41	6.64	8.60	7.29	7.66	7.37
A-Mem	**19.28**	**14.69**	**16.65**	**12.23**	**11.85**	**9.61**	**34.72**	**30.05**	**35.99**	**34.87**
Claude 3.5 Haiku										
LoCoMo	11.34	8.21	3.29	2.69	3.79	3.58	14.01	12.57	7.37	7.12
MemGPT	8.27	6.55	3.99	2.76	4.71	4.48	16.52	14.89	5.64	5.45
A-Mem	**29.70**	**23.19**	**31.54**	**27.53**	**11.42**	**9.47**	**42.60**	**37.41**	**13.65**	**12.71**

These visualizations validate that A-MEM can autonomously maintain meaningful memory structures through its dynamic evolution and linking mechanisms.

A.5 Hyperparameters setting

All hyperparameter k values are presented in Table 8. For models that have already achieved state-of-the-art (SOTA) performance with k=10, we maintain this value without further tuning.

表6： 在LoCoMo数据集上针对问答任务（多跳、时序、开放域、单跳和对抗）五种类别的实验结果，使用不同方法。结果以METEOR和SBERT相似度分数（缩写为ME和SBERT）报告，最佳性能以粗体标出，我们提出的方法A-MEM（以灰色突出显示）在六种基础语言模型上表现出具有竞争力的性能。

模型		方法	类别									
			多跳		时序		开放域		单跳		对抗	
			记忆进化	SBERT	记忆进化	SBERT	记忆进化	SBERT	记忆进化	SBERT	记忆进化	SBERT
GPT	4o-mini	LOCOMO	15.81	47.97	7.61	52.30	8.16	35.00	40.42	57.78	**63.28**	**71.93**
		读取代理	5.46	28.67	4.76	45.07	3.69	26.72	8.01	26.78	8.38	15.20
		记忆库	3.42	21.71	4.07	37.58	4.21	23.71	5.81	20.76	6.24	13.00
		MEMGPT	15.79	49.33	13.25	61.53	4.59	32.77	41.40	58.19	39.16	47.24
		A-MEM	**16.36**	**49.46**	**23.43**	**70.49**	**8.36**	**38.48**	**42.32**	**59.38**	45.64	53.26
	4o	LOCOMO	16.34	53.82	7.21	32.15	8.98	**43.72**	**53.39**	**73.40**	**47.72**	**56.09**
		READAGENT	7.86	37.41	3.76	26.22	4.42	30.75	9.36	31.37	5.47	12.34
		MEMORYBANK	3.22	26.23	2.29	23.49	4.18	24.89	6.64	23.90	2.93	10.01
		MEMGPT	16.64	55.12	12.68	35.93	7.78	37.91	52.14	72.83	31.15	39.08
		A-MEM	**17.53**	**55.96**	**13.10**	**45.40**	**10.62**	38.87	41.93	62.47	32.34	40.11
Qwen2.5	1.5b	LOCOMO	4.99	32.23	2.86	34.03	5.89	35.61	8.57	29.47	40.53	50.49
		READAGENT	3.67	28.20	1.88	27.27	8.97	35.13	5.52	26.33	24.04	34.12
		MEMORYBANK	5.57	35.40	2.80	32.47	4.27	33.85	10.59	32.16	32.93	42.83
		MEMGPT	5.40	35.64	2.35	39.04	7.68	40.36	7.07	30.16	27.24	40.63
		A-MEM	**9.49**	**43.49**	**11.92**	**61.65**	**9.11**	**42.58**	**19.69**	**41.93**	**40.64**	**52.44**
	3b	LOCOMO	2.00	24.37	1.92	25.24	3.45	25.38	6.00	21.28	16.67	23.14
		READAGENT	1.78	21.10	1.69	20.78	4.43	25.15	3.37	18.20	10.46	17.39
		MEMORYBANK	2.37	17.81	2.22	21.93	3.86	20.65	3.99	16.26	15.49	20.77
		MEMGPT	3.74	24.31	2.25	27.67	6.44	29.59	6.24	22.40	13.19	20.83
		A-MEM	**6.25**	**33.72**	**14.04**	**62.54**	**6.56**	**30.60**	**15.98**	**33.98**	**27.36**	**33.72**
Llama 3.2	1b	LOCOMO	5.77	38.02	3.38	45.44	6.20	42.69	9.33	34.19	46.79	60.74
		READAGENT	2.97	29.26	1.31	26.45	7.13	39.19	5.36	26.44	42.39	54.35
		MEMORYBANK	6.77	39.33	4.43	45.63	7.76	42.81	13.01	37.32	50.43	60.81
		MEMGPT	5.10	32.99	2.54	41.81	3.26	35.99	6.62	30.68	45.00	61.33
		A-MEM	**9.01**	**45.16**	**7.50**	**54.79**	**8.30**	**43.42**	**22.46**	**47.07**	**53.72**	**68.00**
	3b	LOCOMO	3.69	27.94	2.96	20.40	6.46	32.17	6.58	22.92	29.02	35.74
		READAGENT	1.21	17.40	2.33	12.02	3.39	19.63	2.46	14.63	14.37	21.25
		MEMORYBANK	3.84	25.06	2.73	13.65	3.05	21.08	6.35	22.02	17.14	24.39
		MEMGPT	2.78	22.06	2.21	14.97	3.63	23.18	3.47	17.81	20.50	26.87
		A-MEM	**9.74**	**39.32**	**13.19**	**59.70**	**8.09**	**32.27**	**24.30**	**42.86**	**39.74**	**46.76**

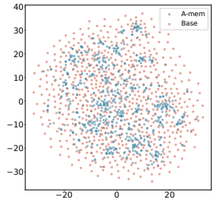
表7： 在LoCoMo数据集上针对问答任务（包含多跳、时序、开放域、单跳和对抗五个类别）使用不同方法进行的实验结果。结果以F1和BLEU-1（%）分数报告，并使用不同的基础模型进行评估。

方法	类别									
	多跳		时序		开放域		单跳		对抗	
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
DeepSeek-R1-32B										
LOCOMO	8.58	6.48	4.79	4.35	12.96	12.52	10.72	8.20	21.40	20.23
MEMGPT	8.28	6.25	5.45	4.97	10.97	9.09	11.34	9.03	**30.77**	**29.23**
A-MEM	**15.02**	**10.64**	**14.64**	**11.01**	**14.81**	**12.82**	**15.37**	**12.30**	27.92	27.19
Claude 3.0 Haiku										
LOCOMO	4.56	3.33	0.82	0.59	2.86	3.22	3.56	3.24	3.46	3.42
MEMGPT	7.65	6.36	1.65	1.26	7.41	6.64	8.60	7.29	7.66	7.37
A-MEM	**19.28**	**14.69**	**16.65**	**12.23**	**11.85**	**9.61**	**34.72**	**30.05**	**35.99**	**34.87**
Claude 3.5 Haiku										
LOCOMO	11.34	8.21	3.29	2.69	3.79	3.58	14.01	12.57	7.37	7.12
MEMGPT	8.27	6.55	3.99	2.76	4.71	4.48	16.52	14.89	5.64	5.45
A-MEM	**29.70**	**23.19**	**31.54**	**27.53**	**11.42**	**9.47**	**42.60**	**37.41**	**13.65**	**12.71**

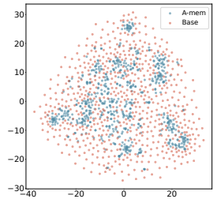
这些可视化验证了A-MEM可以通过其动态进化和链接机制自主地维持有意义的记忆结构。

A.5 超参数设置

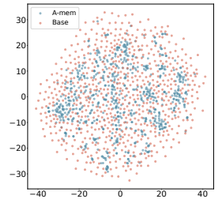
所有超参数k值均展示在表8中。对于已通过k=10实现最先进（SOTA）性能的模型，我们保持此值不再进行调优。



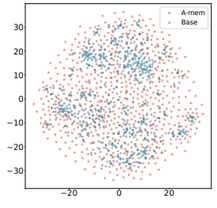
(a) Dialogue 3



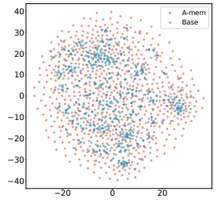
(b) Dialogue 4



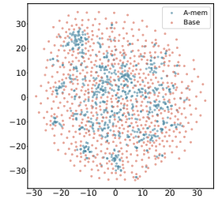
(c) Dialogue 5



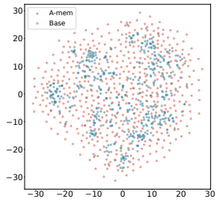
(d) Dialogue 6



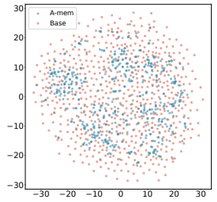
(e) Dialogue 7



(f) Dialogue 8



(g) Dialogue 9

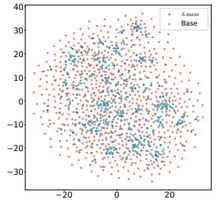


(h) Dialogue 10

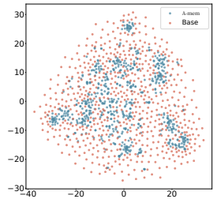
Figure 5: T-SNE Visualization of Memory Embeddings Showing More Organized Distribution with A-MEM (blue) Compared to Base Memory (red) Across Different Dialogues. Base Memory represents A-MEM without link generation and memory evolution.

Table 8: Selection of k values in retriever across specific categories and model choices.

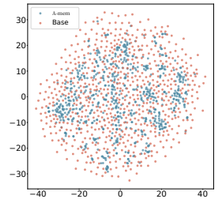
Model	Multi Hop	Temporal	Open Domain	Single Hop	Adversial
GPT-4o-mini	40	40	50	50	40
GPT-4o	40	40	50	50	40
Qwen2.5-1.5b	10	10	10	10	10
Qwen2.5-3b	10	10	50	10	10
Llama3.2-1b	10	10	10	10	10
Llama3.2-3b	10	20	10	10	10



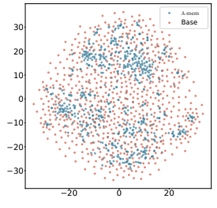
(a) 对话 3



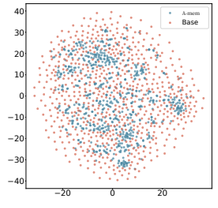
(b) 对话 4



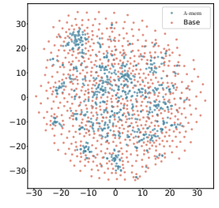
(c) 对话 5



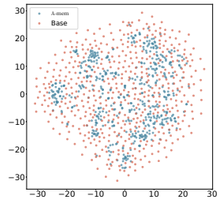
(d) 对话 6



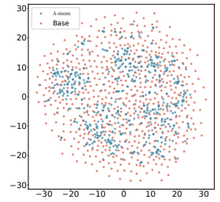
(e)对话 7



(f)对话 8



(g)对话 9



(h) 对话 10

图5： t-SNE 可视化记忆嵌入，显示 A-MEM（蓝色）与基础记忆（红色）在不同对话中的分布更组织化。基础记忆代表没有链接生成和记忆进化的 A-MEM。

表8： 检索器中特定类别和模型选择下 k 值的选择。

模型	多跳	时序	开放域	单跳	对抗
GPT-4o-mini	40	40	50	50	40
GPT-4o	40	40	50	50	40
Qwen2.5-1.5b	10	10	10	10	10
Qwen2.5-3b	10	10	50	10	10
Llama3.2-1b	10	10	10	10	10
Llama3.2-3b	10	20	10	10	10

B Prompt Templates and Examples

B.1 Prompt Template of Note Construction

The prompt template in Note Construction: P_{s1}

```
Generate a structured analysis of the following content by:
1. Identifying the most salient keywords (focus on nouns, verbs, and
key concepts)
2. Extracting core themes and contextual elements
3. Creating relevant categorical tags
Format the response as a JSON object:
{
"keywords": [ // several specific, distinct keywords that capture
key concepts and terminology // Order from most to least important //
Don't include keywords that are the name of the speaker or time // At
least three keywords, but don't be too redundant. ],
"context": // one sentence summarizing: // - Main topic/domain // -
Key arguments/points // - Intended audience/purpose ,
"tags": [ // several broad categories/themes for classification //
Include domain, format, and type tags // At least three tags, but
don't be too redundant. ]
}
Content for analysis:
```

B.2 Prompt Template of Link Generation

The prompt template in Link Generation: P_{s2}

```
You are an AI memory evolution agent responsible for managing and
evolving a knowledge base.
Analyze the the new memory note according to keywords and context,
also with their several nearest neighbors memory.
The new memory context:
{context} content: {content}
keywords: {keywords}
The nearest neighbors memories: {nearest_neighbors_memories}
Based on this information, determine:
Should this memory be evolved? Consider its relationships with other
memories.
```


B 提示模板和示例

B.1 笔记构建提示模板

笔记构建中的提示模板： P_{s1}

```
通过以下方式生成结构化分析：
1. 识别最突出的关键词（关注名词、动词和
关键概念）
2. 提取核心主题和上下文元素
3. 创建相关的分类标签
将响应格式化为 JSON 对象：
{
"keywords": [ // 几个具体且不同的关键词，用于捕捉
关键概念和术语 // 按重要性从高到低排序 //
不要包含说话者名称或时间等关键词 // At
至少三个关键词，但不要过于冗余。 ],
"上下文": // 一句总结: // - 主要主题/领域 // -
关键论点/要点 // - 目标受众/目的 ,
"标签": [ // 用于分类的几个广泛类别/主题 //
包含领域、格式和类型标签 // 至少三个标签，但
不要过于冗余。 ]
}
分析内容：
```

B.2 链接生成提示模板

链接生成中的提示模板： P_{s2}

```
你是负责管理和进化知识库的AI记忆进化代理。
进化知识库。
根据关键词和上下文分析新的记忆笔记，
同时结合其最近的几个邻居记忆。
新的记忆上下文：
{context}内容： {content}
关键词： {keywords}
最近邻记忆： {最近_邻居_记忆}
根据这些信息 ，确定： 考虑它与其他
这个记忆是否应该进化
记忆的关系
```

B.3 Prompt Template of Memory Evolution

The prompt template in Memory Evolution: P_{s3}

```
You are an AI memory evolution agent responsible for managing and
evolving a knowledge base.
Analyze the the new memory note according to keywords and context,
also with their several nearest neighbors memory.
Make decisions about its evolution.
The new memory context:{context}
content: {content}
keywords: {keywords}
The nearest neighbors memories:{nearest_neighbors_memories}
Based on this information, determine:
1. What specific actions should be taken (strengthen,
update_neighbor)?
1.1 If choose to strengthen the connection, which memory should it be
connected to? Can you give the updated tags of this memory?
1.2 If choose to update neighbor, you can update the context and tags
of these memories based on the understanding of these memories.
Tags should be determined by the content of these characteristic
of these memories, which can be used to retrieve them later and
categorize them.
All the above information should be returned in a list format
according to the sequence: [[new_memory],[neighbor_memory_1],
...[neighbor_memory_n]]
These actions can be combined.
Return your decision in JSON format with the following structure: {{
"should_evolve": true/false,
"actions": ["strengthen", "merge", "prune"],
"suggested_connections": ["neighbor_memory_ids"],
"tags_to_update": ["tag_1",..."tag_n"],
"new_context_neighborhood": ["new context",...,"new context"],
"new_tags_neighborhood": [["tag_1",...,"tag_n"],...["tag_1",...,"tag_n"]],
}}
```


B.3 记忆进化提示模板

记忆进化中的提示模板： P_{s3}

```
你是负责管理和
进化知识库的人工智能记忆进化代理。
根据关键词和上下文分析新的记忆笔记，
同时结合其最近的几个邻居记忆。
就其进化做出决策。
新的记忆上下文:{context}
内容: {content}
关键词: {keywords}
最近邻记忆:{nearest_neighbors_memories}
基于这些信息，确定:
1. 应该采取哪些具体动作（强化、
更新_邻居）？
1.1 如果选择强化连接，应该选择哪个记忆
连接到？你能给出这个记忆的更新标签吗？
1.2 如果选择更新邻居，你可以更新上下文和标签
根据对这些记忆的理解，更新这些记忆的标签。
标签应由这些记忆的特征内容决定
这些记忆的特征，可用于稍后检索它们，并
对它们进行分类。
以上所有信息应以列表格式返回
按照以下顺序: [[新_记忆],[邻居_记忆_1],
...[邻居_记忆_n]]
这些动作可以组合。
以 JSON 格式返回您的决策，并使用以下结构: {{
"should_进化": true/false，
"actions": ["强化", "合并", "剪枝"],
"建议_连接": ["邻居_记忆_ids"],
"标签_要_更新": ["标签_1",..."标签_n"],
"新_上下文_邻域":, ["新上下文",...,"新上下文"],
"新_标签_邻域":, [["标签_1",...,"标签_n"],...["标签_1",...,"标签_n"]],
}}
```

B.4 Examples of Q/A with A-MEM

Example:
Question 686: Which hobby did Dave pick up in October 2023?
Prediction: photography
Reference: photography
talk start time:10:54 am on 17 November, 2023
memory content: Speaker Davesays : Hey Calvin, long time no talk! A lot has happened. I've taken up photography and it's been great - been taking pics of the scenery around here which is really cool.
memory context: The main topic is the speaker's new hobby of photography, highlighting their enjoyment of capturing local scenery, aimed at engaging a friend in conversation about personal experiences.
memory keywords: ['photography', 'scenery', 'conversation', 'experience', 'hobby']
memory tags: ['hobby', 'photography', 'personal development', 'conversation', 'leisure']
talk start time:6:38 pm on 21 July, 2023
memory content: Speaker Calvinsays : Thanks, Dave! It feels great having my own space to work in. I've been experimenting with different genres lately, pushing myself out of my comfort zone. Adding electronic elements to my songs gives them a fresh vibe. It's been an exciting process of self-discovery and growth!
memory context: The speaker discusses their creative process in music, highlighting experimentation with genres and the incorporation of electronic elements for personal growth and artistic evolution.
memory keywords: ['space', 'experimentation', 'genres', 'electronic', 'self-discovery', 'growth']
memory tags: ['music', 'creativity', 'self-improvement', 'artistic expression']

B.4 A-MEM的问答示例

示例:
问题686: Dave在2023年10月养成了什么爱好?
预测: 摄影
参考文献: 摄影
演讲开始时间: 上午10:54, 11月17日 , 2023
记忆内容: 发言人Davesays: 嘿Calvin, 好久不见! 发生了很多事。 我开始摄影了, 这很棒——一直在拍这里的风景照片, 这真的很酷。
记忆上下文: 主要话题是说话者新的爱好, 摄影, 强调他们捕捉当地风景, 旨在与朋友就个人经历展开对话。
记忆关键词: ['摄影', '风景', '对话', '经历' '爱好']
记忆标签: ['爱好', '摄影', '个人成长', '对话', '休闲']
谈话开始时间: 2023年7月21日下午6:38
记忆内容: 说话人Calvin说: 谢谢, Dave! 感觉有自己的空间来工作真是太棒了。 我一直都在尝试不同的风格, 努力突破自己的舒适区。 给我的歌曲中加入电子元素, 让它们焕发新鲜感。 It's 这是一个令人兴奋的自我发现和成长的过程!
记忆上下文: 演讲者讨论了他们在音乐中的创作过程, 强调了跨流派实验以及电子元素的融合, 以促进个人成长和艺术进化。
记忆关键词: ['空间', '实验', '流派', '电子', '自我发现', '成长']
记忆标签: ['音乐', '创造力', '自我提升', '艺术表达']

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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• 如果贡献是一个数据集和/或模型，作者应该描述他们为使结果可复现或可验证所采取的步骤。
• 根据贡献的性质，可复现性可以通过多种方式实现。例如，如果贡献是一个新架构，充分描述该架构可能就足够了；如果贡献是一个特定模型和实证评估，可能需要让其他人使用相同的数据集来复现该模型，或者提供模型访问权限。通常来说，发布代码和数据是实现这一点的一种好方法，但可复现性也可以通过提供详细的复现结果说明、访问托管模型（例如，对于大型语言模型的情况）、发布模型检查点或其他适合所进行研究的手段来提供。

• 虽然NeurIPS不要求发布代码，但该会议要求所有提交的论文提供某种合理的可复现性途径，这可能取决于贡献的性质。例如（a）如果贡献主要是新算法，论文应明确说明如何复现该算法。（b）如果贡献主要是新模型架构，论文应清晰地全面描述该架构。（c）如果贡献是新模型（例如，大型语言模型），则应有访问此模型以复现结果的方式，或复现模型的方式（例如，使用开源数据集或构建数据集的说明）。（d）我们认识到在某些情况下可复现性可能很复杂，在这种情况下，作者可以描述他们提供的特定可复现性方式。对于闭源模型，模型访问可能以某种方式受限（例如，仅限注册用户），但其他研究人员应该有可能通过某种途径来复现或验证结果。

5. **开放数据与代码访问**

问题：论文是否提供开放的数据和代码，并附有足够的说明以忠实地复现补充材料中描述的主要实验结果?

回答： [是]

Ju说明：我们在摘要中提供了代码链接 act.

指南：
• 答案NA表示该论文不包含需要代码的实验。
• 请参阅NeurIPS代码和数据提交指南（https://nips.cc/public/guides/CodeSubmissionPolicy）以获取更多详细信息。
• 虽然我们鼓励发布代码和数据，但我们理解这可能并不可行，因此“否”也是一个可接受的答案。论文不能仅仅因为未包含代码而被拒绝，除非这对其贡献至关重要（例如，用于一个新的开源基准）。

• 指令应包含运行以复现结果的精确命令和环境。请参阅NeurIPS代码和数据提交指南（https://nips.cc/public/guides/CodeSubmissionPolicy）以获取更多详细信息。
• 作者应提供数据访问和准备说明，包括如何访问原始数据、预处理数据、中间数据和生成数据等。
• 作者应提供脚本以复现新提出的方法和基线的所有实验结果。如果只有部分实验可复现，他们应说明哪些实验从脚本中省略了以及原因。

• 提交时，为保护匿名性，作者应发布匿名版本（如适用）。

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: We cover all the details in the paper.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: The experiments utilize the API of Large Language Models. Multiple calls will significantly increase costs.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
   - If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: It could be found in the experimental part.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- 建议在补充材料（附加到论文中）中尽可能提供更多信息，但允许包含数据代码的URL。

6. **实验设置/详情**

   问题：论文是否详细说明了所有理解结果所需的训练和测试细节（例如，数据划分、超参数、如何选择、优化器类型等）？

   回答： [是]

   理由：我们在论文中涵盖了所有细节。

   指南：

   - 回答NA表示该论文未包含实验。
   - 实验设置应在论文核心部分呈现，详细程度需足以理解结果并对其进行分析。
   - 完整细节可通过代码、附录或补充材料提供。

7. **实验统计显著性**

   问题：论文是否适当且正确地报告了误差线或其他关于实验统计显著性的信息？

   回答： [否]

   理由：实验使用大型语言模型的API。多次调用将显著增加成本。

   指南：

   - 答案为NA表示该论文不包含实验。
   - 如果结果伴有误差线、置信区间或统计显著性检验（至少针对支持论文主要论点的实验），作者应回答“是”。
   - 误差线所捕捉的变异因素应明确说明（例如，训练/测试分割、初始化、某些参数的随机抽取，或给定实验条件下的整体运行）。
   - 计算误差线的方法应予以解释（封闭形式公式、调用库函数、自助法等）。
   - 所做出的假设应给出（例如，误差正态分布）。
   - 应明确误差线是标准差还是均值的标准误差。
   - 报告1-σ误差线是可以的，但需要说明。如果误差的正态性假设未验证，作者最好报告2-σ误差线，而不是说明他们有96%置信区间。
   - 对于非对称分布，作者应小心不要在表格或图中展示对称误差线，以免导致结果超出范围（例如负误差率）。
   - 如果表格或图中报告了误差线，作者应在文中解释其计算方法，并引用相应的图表或表格。

8. **实验计算资源**

   问题：对于每个实验，论文是否提供了足够的信息来复现实验所需的计算资源（计算工作类型、内存、执行时间）？

   答案： [是]

   理由：可以在实验部分找到。

   指南：

   - 答案NA表示该论文不包含实验。
   - 论文应标明计算工作者的类型，包括CPU或GPU、内部集群或云服务提供商，以及相关的内存和存储。

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://neurips.cc/public/EthicsGuidelines?

Answer: [NA]

Justification: N/A

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: We don't discuss this aspect because we provide only the memory system for LLM agents. Different LLM agents may create varying societal impacts, which are beyond the scope of our work.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: N/A

- 论文应提供每个独立实验运行所需的计算量，并估计总计算量。
- 论文应披露整个研究项目是否需要比论文中报告的实验更多的计算资源（例如，未纳入论文的初步或失败的实验）。

9. **伦理准则**

问题：论文中开展的研究是否在所有方面都符合NeurIPS伦理准则 https://neurips.cc/public/EthicsGuidelines?

回答： [NA]

理由：不适用

指南：
- 答案为NA表示作者尚未审查NeurIPS伦理准则。
- 如果作者回答否，则应解释需要偏离伦理准则的特殊情况。
- 作者应确保保留匿名性（例如，如果由于当地法律或法规存在特殊考虑）。

10. **更广泛的影响**

问题：论文是否讨论了所进行工作的潜在积极社会影响和消极社会影响?

答案： [否]

理由：我们不讨论这个方面，因为我们只提供大语言模型代理的记忆系统。不同的大语言模型代理可能会产生不同的社会影响，这超出了我们工作的范围。

指南：
- 答案为NA表示该工作没有产生社会影响。
- 如果作者回答NA或否，他们应该解释为什么他们的工作没有社会影响，或者为什么论文没有涉及社会影响。
- 负面的社会影响示例包括潜在的恶意或非预期使用（例如，虚假信息、生成虚假资料、监控）、公平性考虑（例如，部署可能做出不公平影响特定群体的决策的技术）、隐私考虑和安全考虑。
- 会议期望许多论文将是基础性研究，并且与特定应用无关，更不用说部署了。然而，如果有任何直接通往负面应用的路径，作者应该指出这一点。例如，指出提高生成模型的质量可能被用于生成虚假信息以传播虚假信息是合理的。另一方面，指出通用的神经网络优化算法可以使人们更快地训练生成深度伪造模型的算法是不必要的。
- 作者应考虑该技术在预期用途且运行正常时可能产生的潜在危害，在技术按预期使用但给出错误结果时可能产生的危害，以及因（故意或无意）滥用该技术而产生的危害。
- 如果存在负面社会影响，作者也可以讨论可能的缓解策略（例如，模型的分阶段发布，提供防御措施而非攻击，监控滥用的机制，监控系统如何随时间从反馈中学习，提高机器学习（ML）的效率和可及性）。

11. **安全防护措施**

问题：论文是否描述了为负责任地发布具有高风险被滥用的数据或模型（例如，预训练语言模型、图像生成器或网络爬取数据集）而采取的安全防护措施?

回答： [不适用]

说明：不适用

Guidelines:

- The answer NA means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Their contribution has already been properly acknowledged and credited.

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
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13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [NA]

Justification: N/A

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

指南：

- 答案NA表示该论文不存在此类风险。
- 对于存在高风险被滥用或具有双重用途的已发布模型，应采取必要的防护措施以允许对模型进行受控使用，例如要求用户遵守使用指南或限制访问模型，或实施安全过滤器。
- 从互联网上收集的数据集可能存在安全风险。作者应说明他们如何避免发布不安全的图像。
- 我们认识到提供有效的防护措施具有挑战性，许多论文并未要求这样做，但我们鼓励作者考虑这一点并尽最大努力。

12. **现有资产许可证**

问题：论文中使用的资产（例如代码、数据、模型）的创建者或原始所有者是否得到了适当的认可，并且许可证和使用条款是否明确提及并得到正确尊重？

答案： [是]

理由：他们的贡献已经被恰当地认可和致谢。

指南：

- 答案NA表示该论文未使用现有资产。
- 作者应该引用产生代码包或数据集的原始论文。
- 作者应该说明使用的资产版本，如果可能的话，请包含一个URL。
- 每个资产的许可证名称（例如，CC-BY 4.0）应包含在内。
- 对于从特定来源（例如，网站）抓取的数据，应提供该来源的版权和服务条款。
- 如果资产被发布，应提供许可证、版权信息和包中的使用条款。对于热门数据集，paperswithcode.com/datasets为一些数据集整理了许可证。他们的许可证指南可以帮助确定数据集的许可证。
- 对于重新打包的现有数据集，应提供原始许可证和派生资产的许可证（如果已更改）。
- 如果此信息在线上不可用，鼓励作者联系资产的创建者。

13. **新资产**

问题：论文中引入的新资产是否得到了充分说明？资产说明是否随资产一同提供？

回答： [NA]

理由：N/A

指南：

- 回答NA表示论文未发布新资产。
- 研究人员应通过结构化模板在提交时沟通数据集/代码/模型的具体细节。这包括关于训练、许可、限制等方面的信息。
- 论文应讨论是否以及如何从使用其资产的人那里获得同意。
- 提交时，请记得匿名化您的资产（如果适用）。您可以创建一个匿名URL，或包含一个匿名压缩文件。

14. **众包与人类受试者研究**

问题：对于众包实验和涉及人类受试者的研究，论文是否包含向参与者提供的完整说明文本以及截图（如适用），以及关于补偿（如有）的详细信息？

Answer: [NA]

Justification: N/A

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: N/A

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigorousness, or originality of the research, declaration is not required.

    Answer: [NA]

    Justification: N/A

    Guidelines:

    - The answer NA means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy (https://neurips.cc/Conferences/2025/LLM) for what should or should not be described.

回答： [不适用]

说明：不适用

指南：

- 答案为NA表示该论文不涉及众包或涉及人类受试者的研究。
- 在补充材料中包含此信息是可以的，但如果论文的主要贡献涉及人类受试者，那么应尽可能在正文中包含详细信息。
- 根据NeurIPSCode of Ethics，参与数据收集、整理或其他劳动的人员应至少获得数据收集所在国家的最低工资。

15. **涉及人类受试者的研究需获得机构审查委员会（IRB）批准或同等批准**

    问题：论文是否描述了研究参与者可能面临的风险，是否向受试者披露了这些风险，以及是否获得了机构审查委员会（IRB）的批准（或基于您国家或机构要求的同等批准/审查）？

    回答： [NA]

    理由：不适用

    指南：

    - 回答NA表示该论文不涉及众包或涉及人类受试者的研究。
    - 根据研究进行的国家，任何涉及人类受试者的研究可能都需要IRB批准（或同等批准）。如果您获得了IRB批准，您应在论文中明确说明。
    - 我们认识到，这项程序在不同机构之间以及不同地点可能会有显著差异，我们期望作者遵守NeurIPS伦理准则和其所在机构的指南。
    - 对于初步提交，不要包含任何可能破坏匿名性（如适用）的信息，例如进行评审的机构。

16. **大语言模型使用声明**

    问题：如果它是本研究核心方法中一个重要、原创或非标准的组成部分，论文是否描述了大语言模型的使用？请注意，如果大语言模型仅用于写作、编辑或格式化目的，并且不影响核心方法、科学严谨性或研究的原创性，则无需声明。

    回答： [NA]

    理由：N/A

    指南：

    - 答案为NA表示，本研究中的核心方法开发不涉及将大语言模型作为任何重要、原创或非标准组件。
    - 请参考我们的<code>大语言模型</code>政策（https://neurips.cc/Conferences/2025/LLM）了解应该或不应该描述的内容。